



Building AI Agents with Multimodal Models

Part 4: NVIDIA AI Blueprint for Video Search & Summarization



Agenda

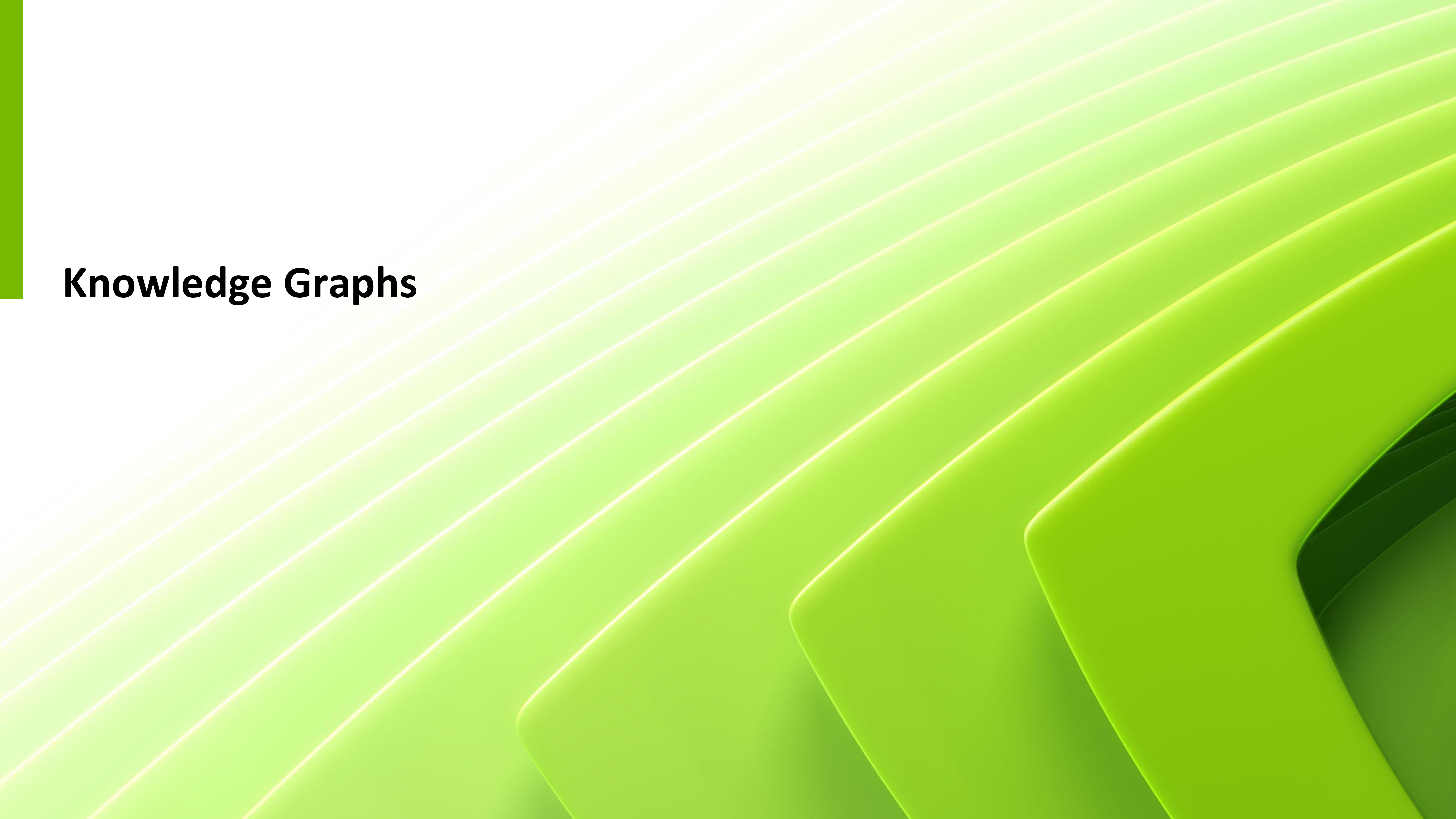
- Part 1: Multimodal Data

- Part 2: Intermediate Fusion

- Part 3: Cross-modal Projection and OCR Pipelines

- Part 4: VSS and GraphRAG

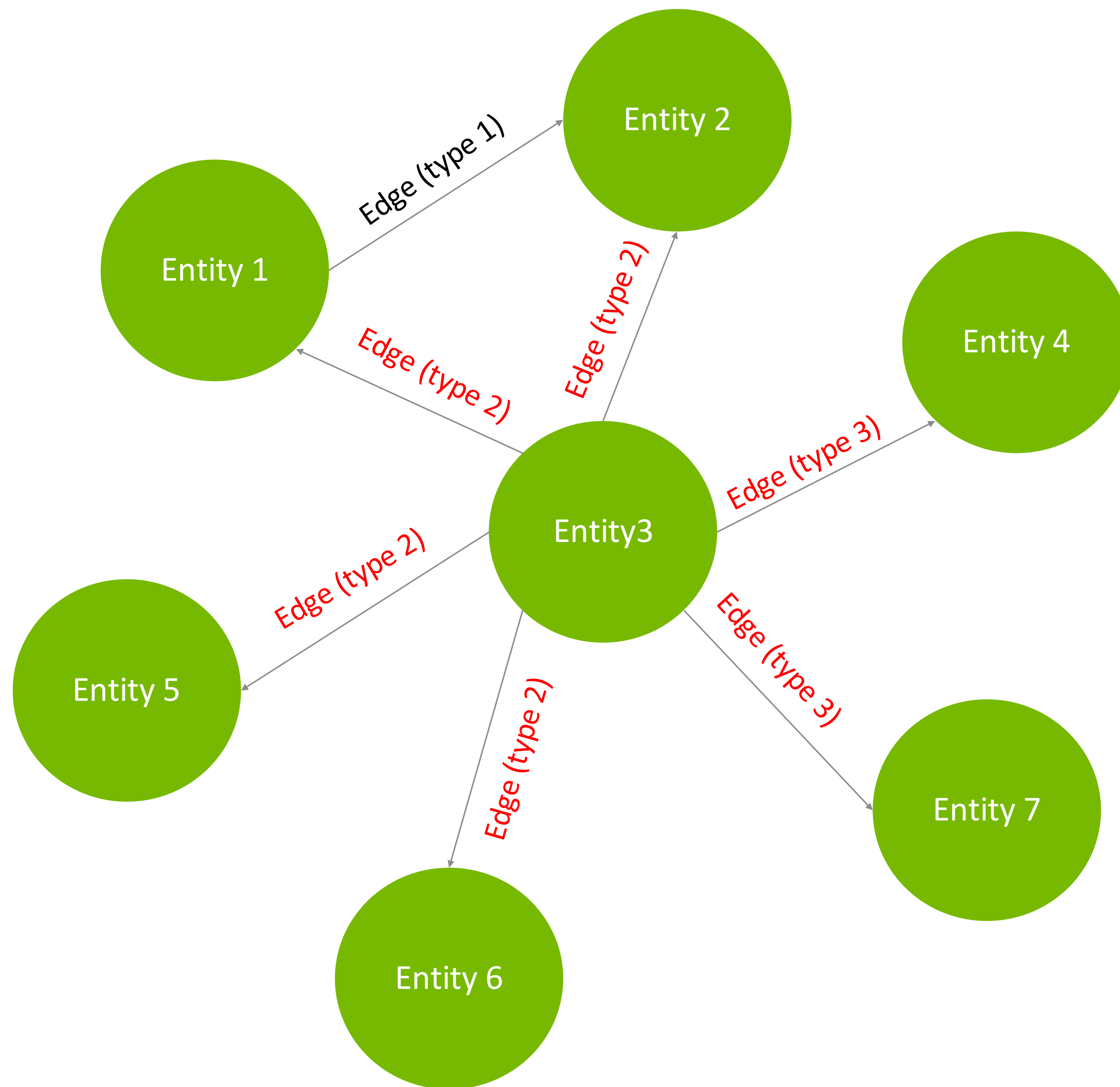
- Part 5: Wrap-up & Assessment



Knowledge Graphs

Knowledge Graph

A refresher

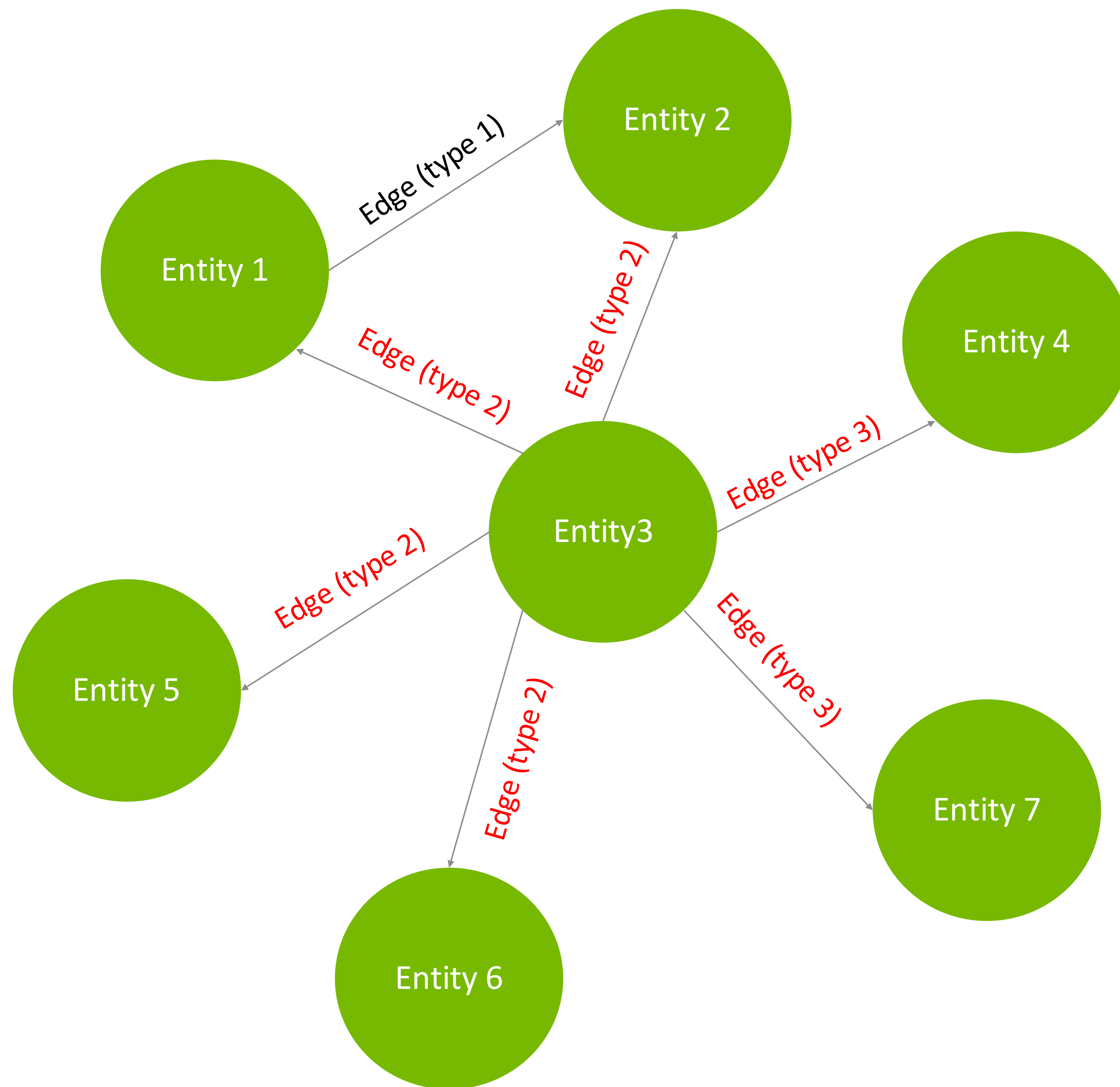


- Knowledge graphs (KGs) are semantic networks that represent entities and the relationships between.
- Entities can be objects, places, people, concepts or events. The edges between them represent the relationship between those entities.
- Two different entities connected by relationships are also called as “Triplets”
- Each node type can have different properties or attributes.
- Each edge or relationship type between the nodes can also have different properties
- Graphs allow for some unique knowledge retrieval capabilities based on its structure.
 - KGs organize data from multiple sources.
 - Capture information about entities for a given domain or task.
 - Forge connections between those entities giving a comprehensive view of the knowledge base.

[['Exxon Mobil', 'COMP', 'Cut', 'Spending on oil and gas exploration', 'ACTIVITY'],

Knowledge Graph

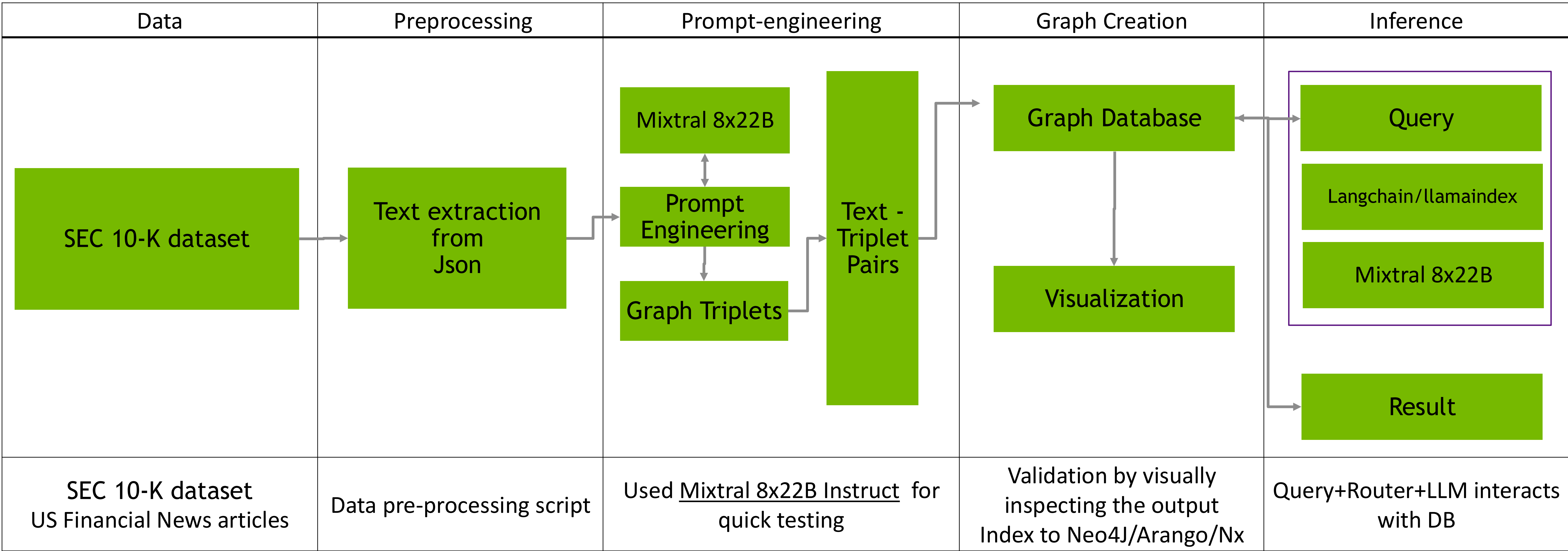
A refresher



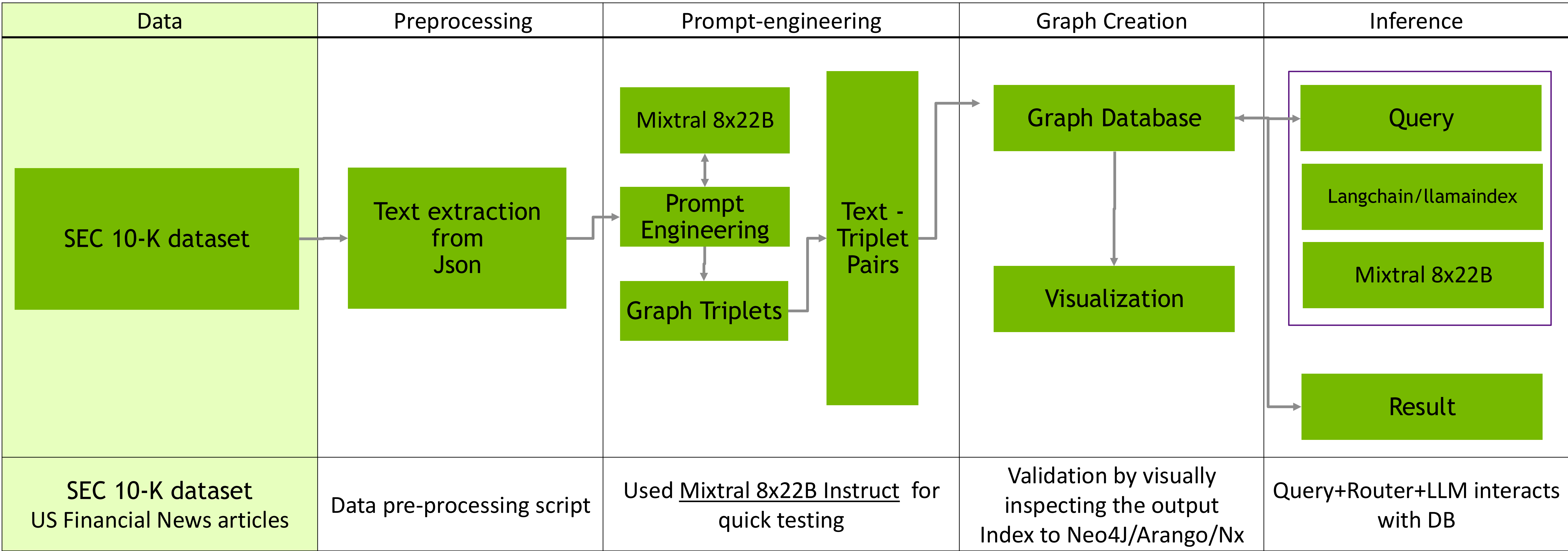
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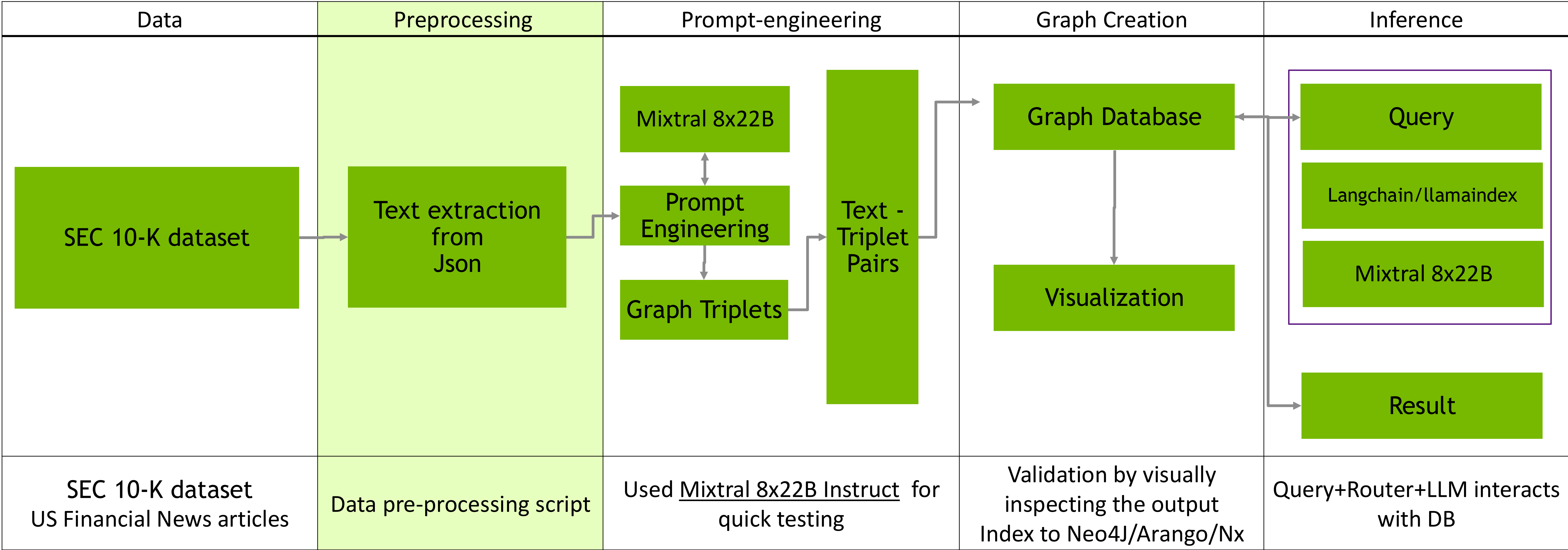

THE OVERALL PROCESS



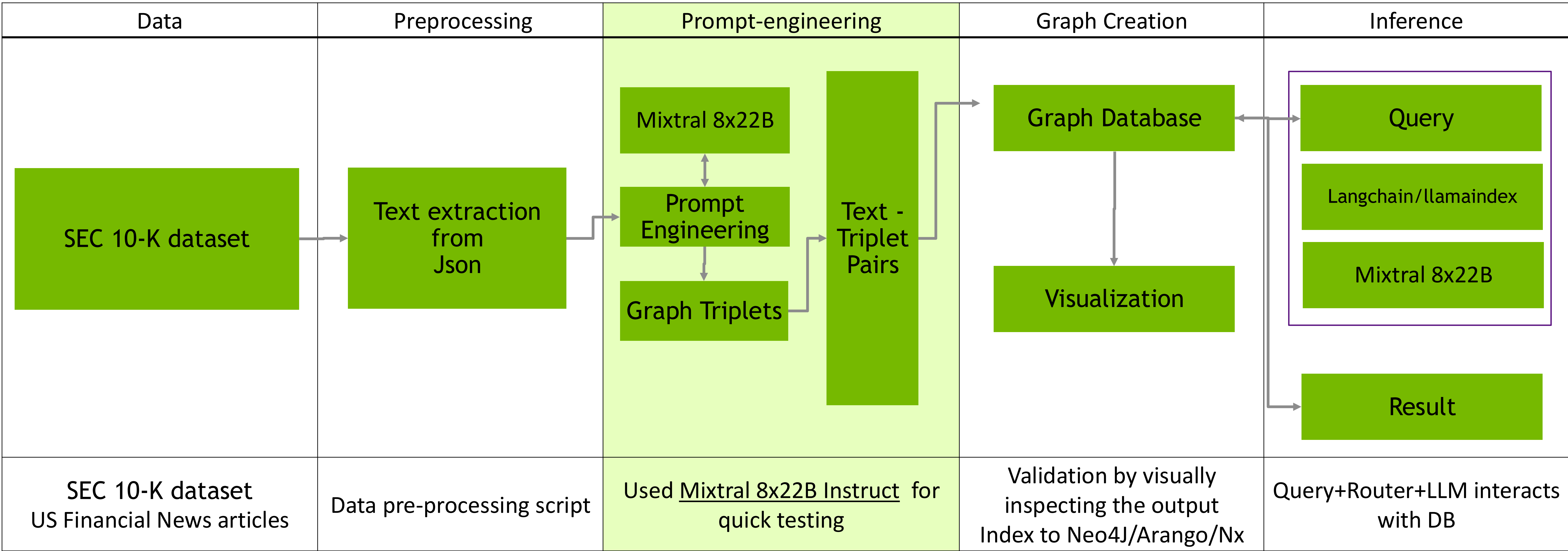
THE OVERALL PROCESS



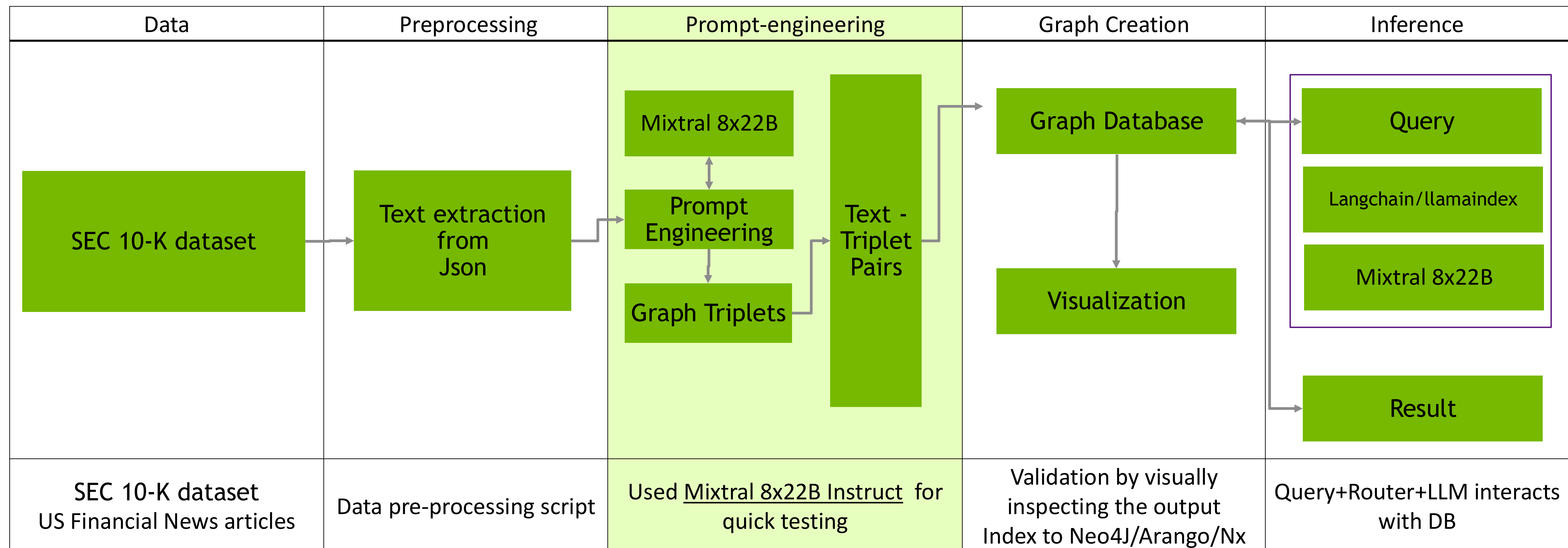
THE OVERALL PROCESS



THE OVERALL PROCESS

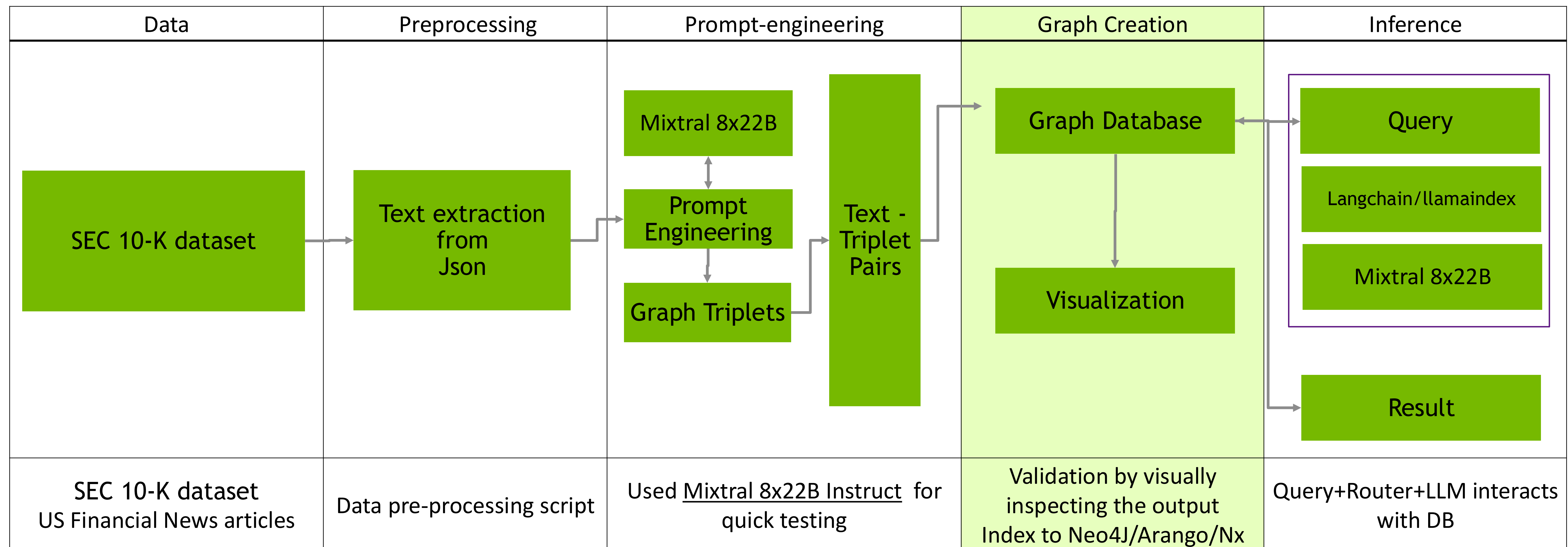


THE OVERALL PROCESS



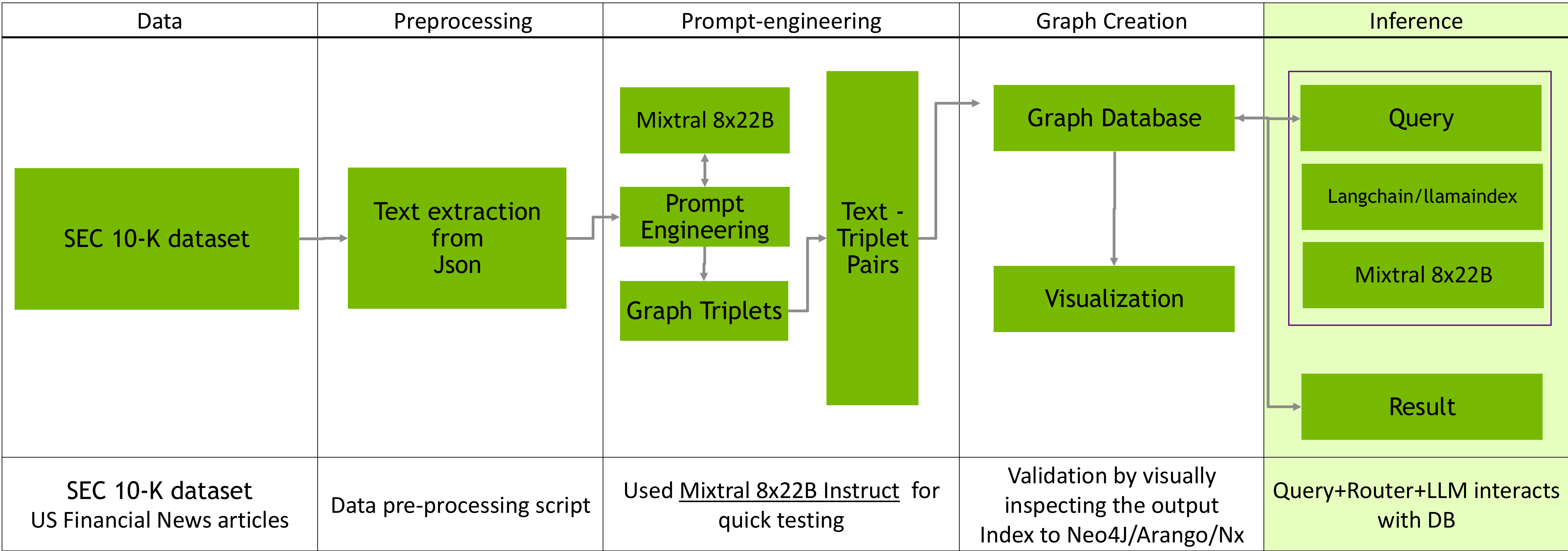
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THE OVERALL PROCESS

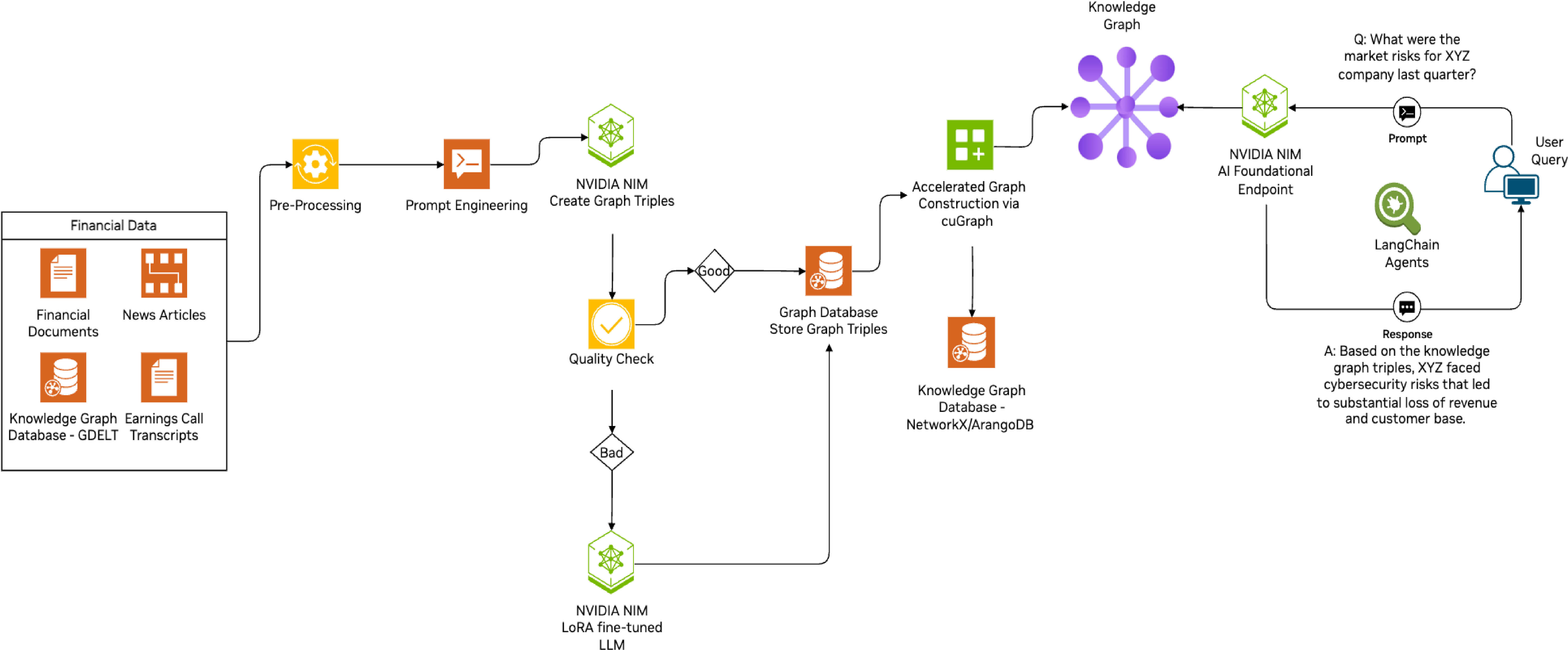


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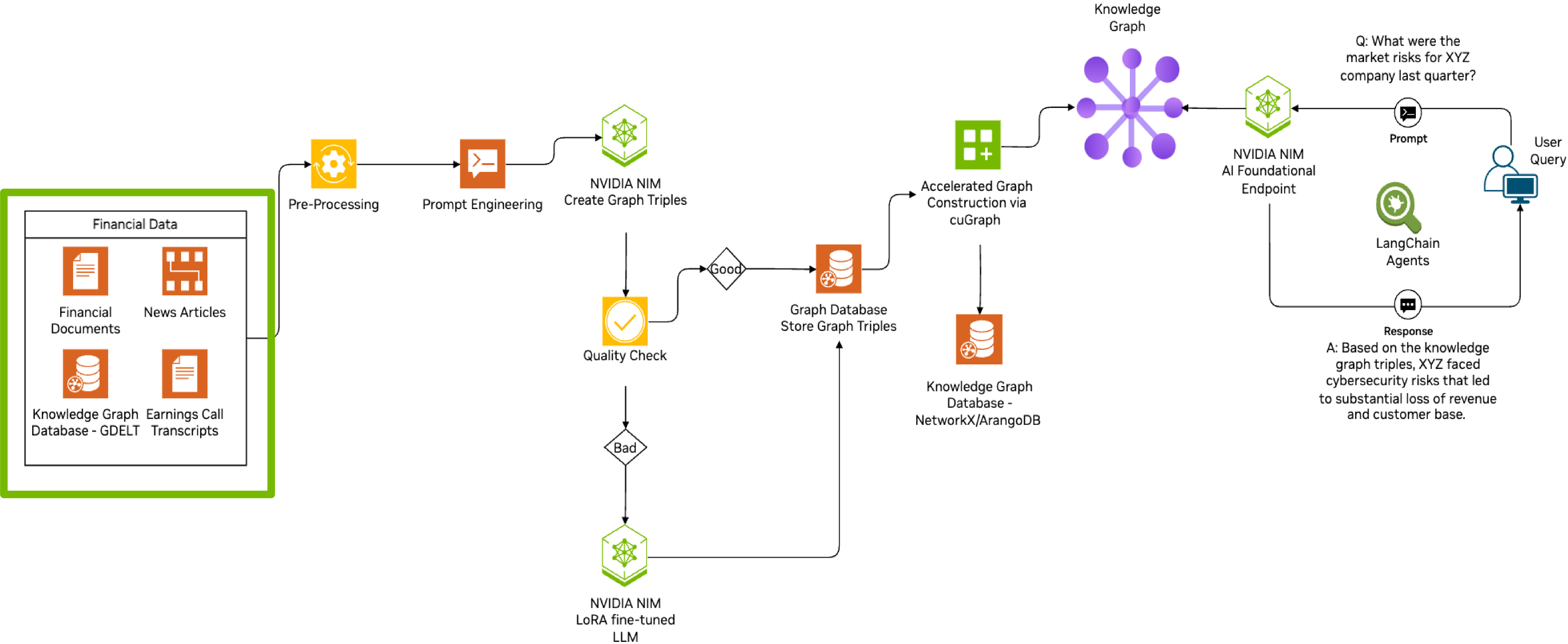

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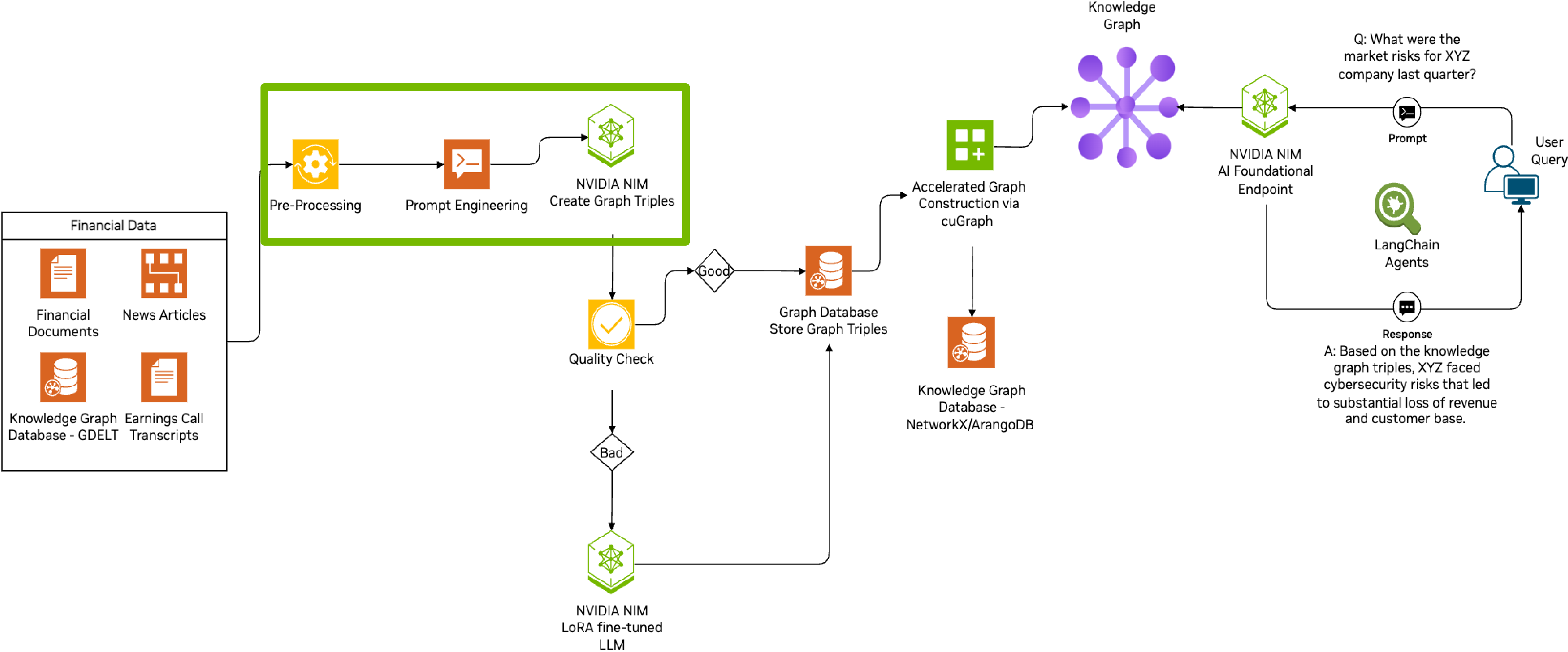
Overall Pipeline and Components



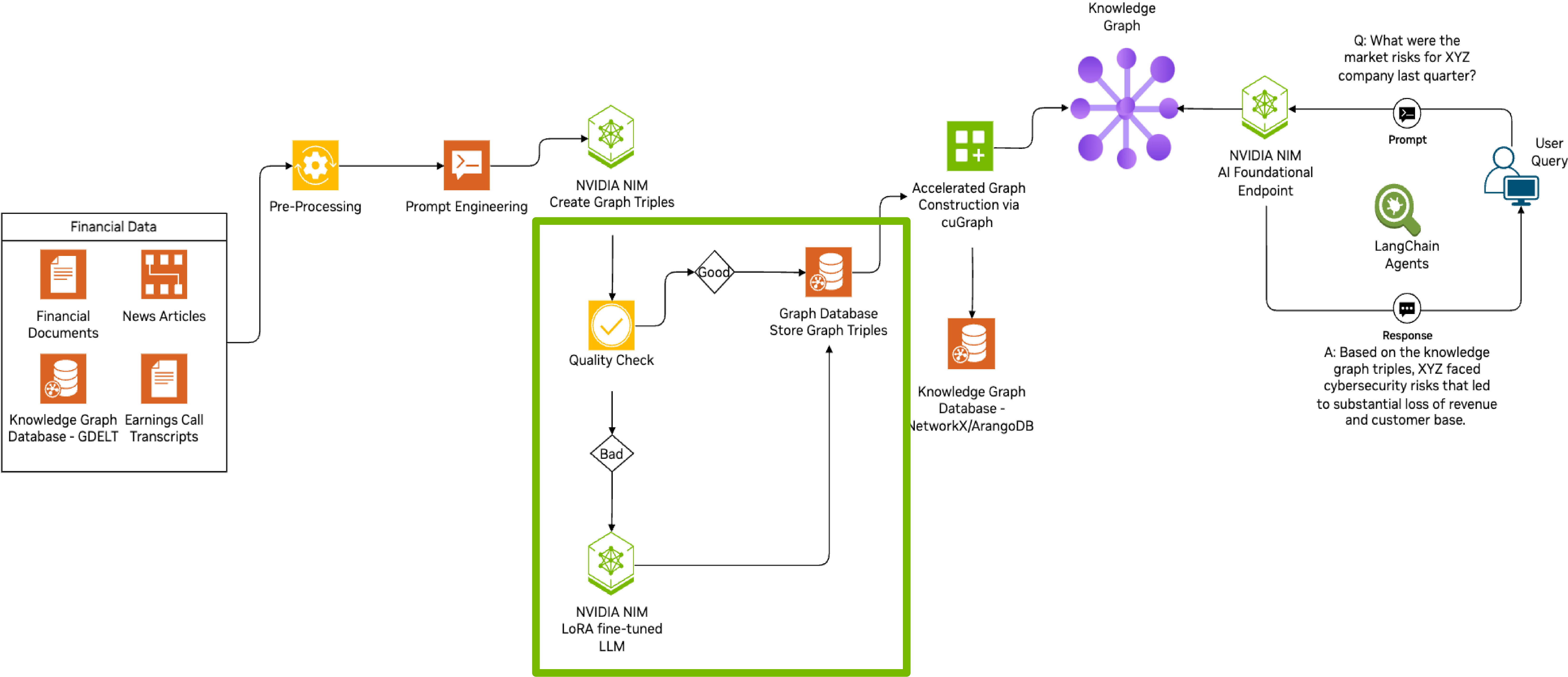
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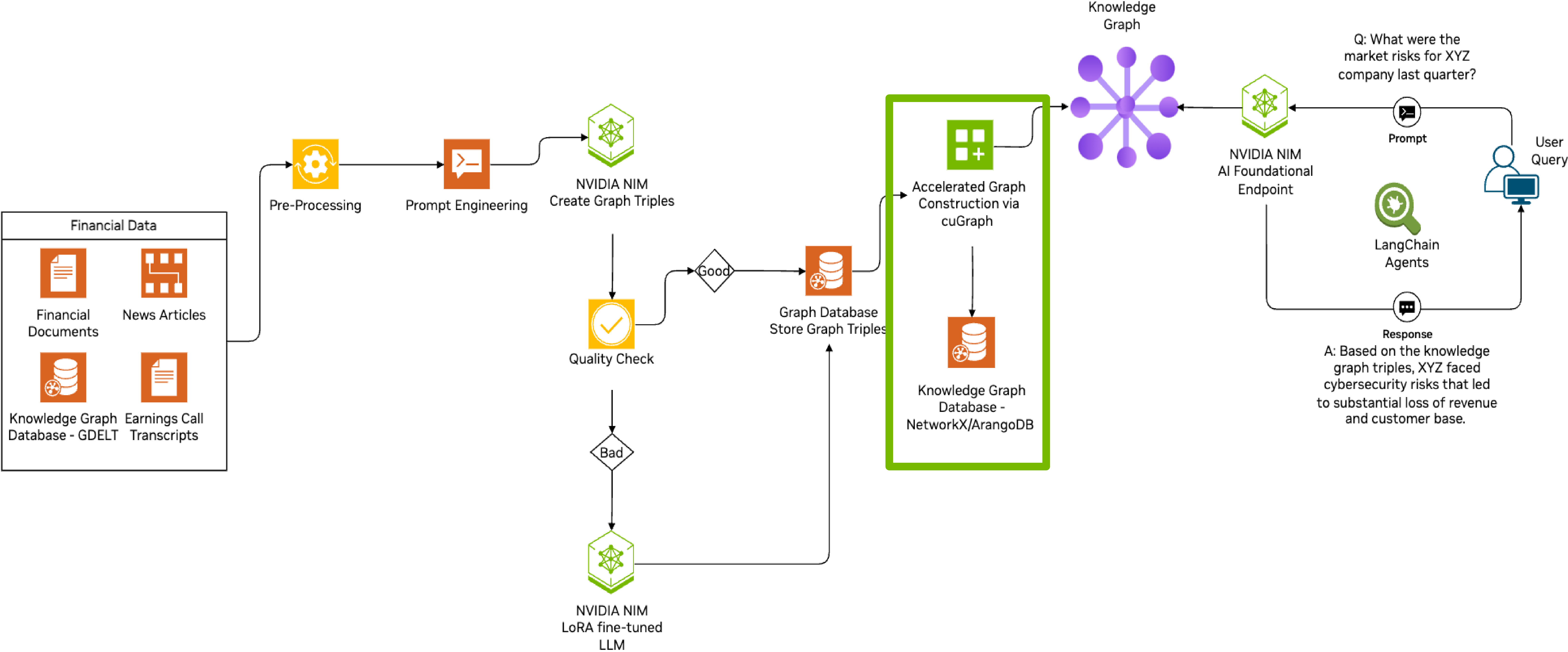
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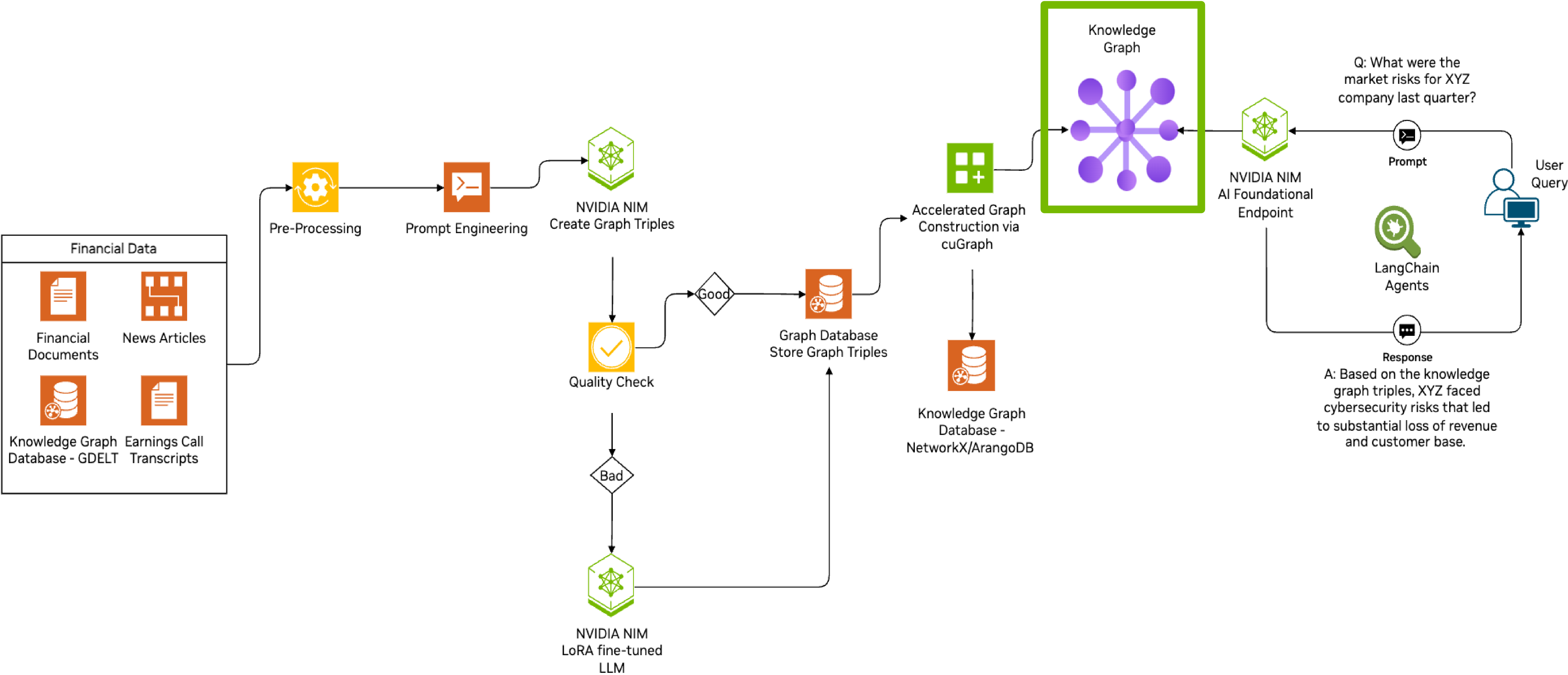
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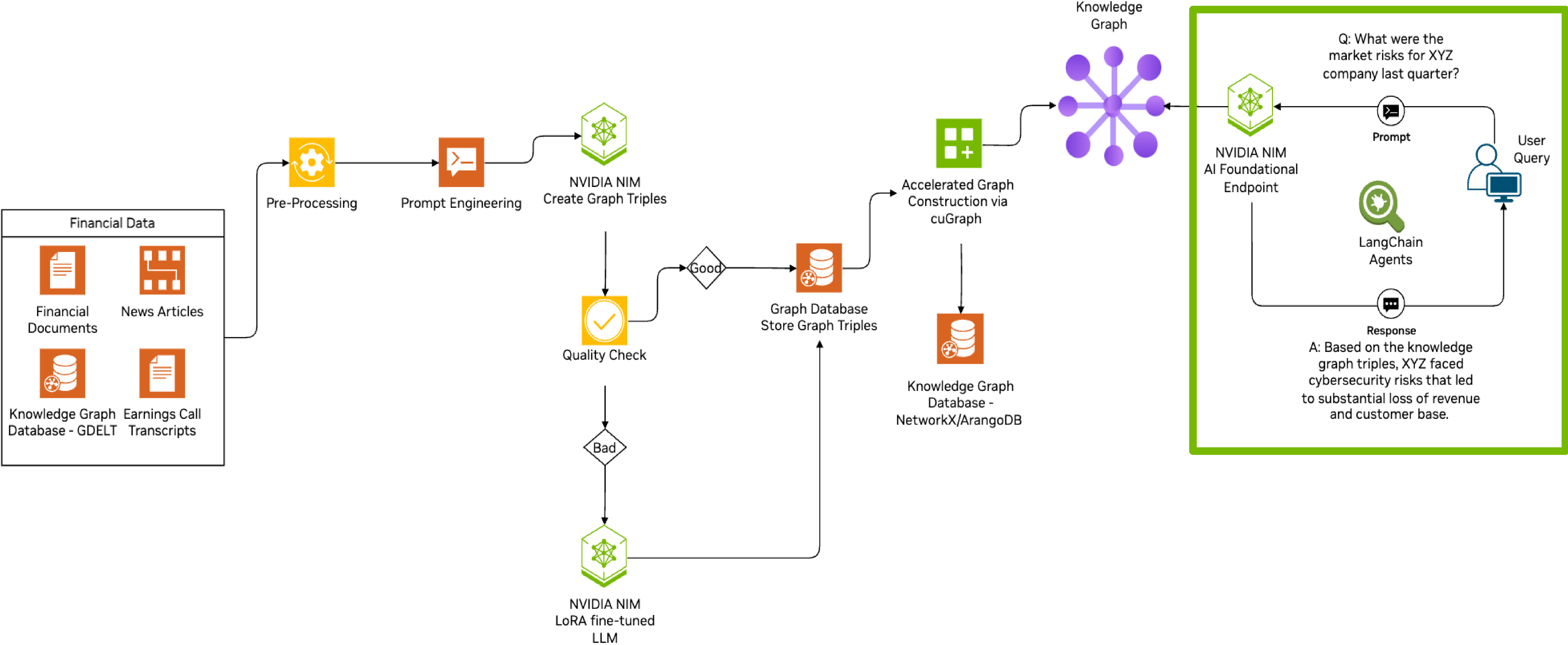
Overall Pipeline and Components



Overall Pipeline and Components



Overall Pipeline and Components



The Triplets

Text

LONDON (Reuters) - Despite the strongest start for oil prices in four years, the world's top oil companies are hesitating to accelerate the search for new resources as a determination to retain capital discipline trumps the hope of making bonanza discoveries.

Exxon Mobil, Royal Dutch Shell, Total and their peers are set to cut spending on oil and gas exploration for a fifth year in a row in 2018, according to consultancy Wood Mackenzie (WoodMac), despite a growing urgency to replenish reserves after years of reining back investment.

(For graphic 'Global spending on oil and gas exploration' click reut.rs/2CjAONv)

Global investment in exploration, vital to increase output and offset the natural decline of existing fields, will reach \$37 billion in 2018, down 7 percent from a year earlier and over 60 percent below the 2014 peak, according to WoodMac.

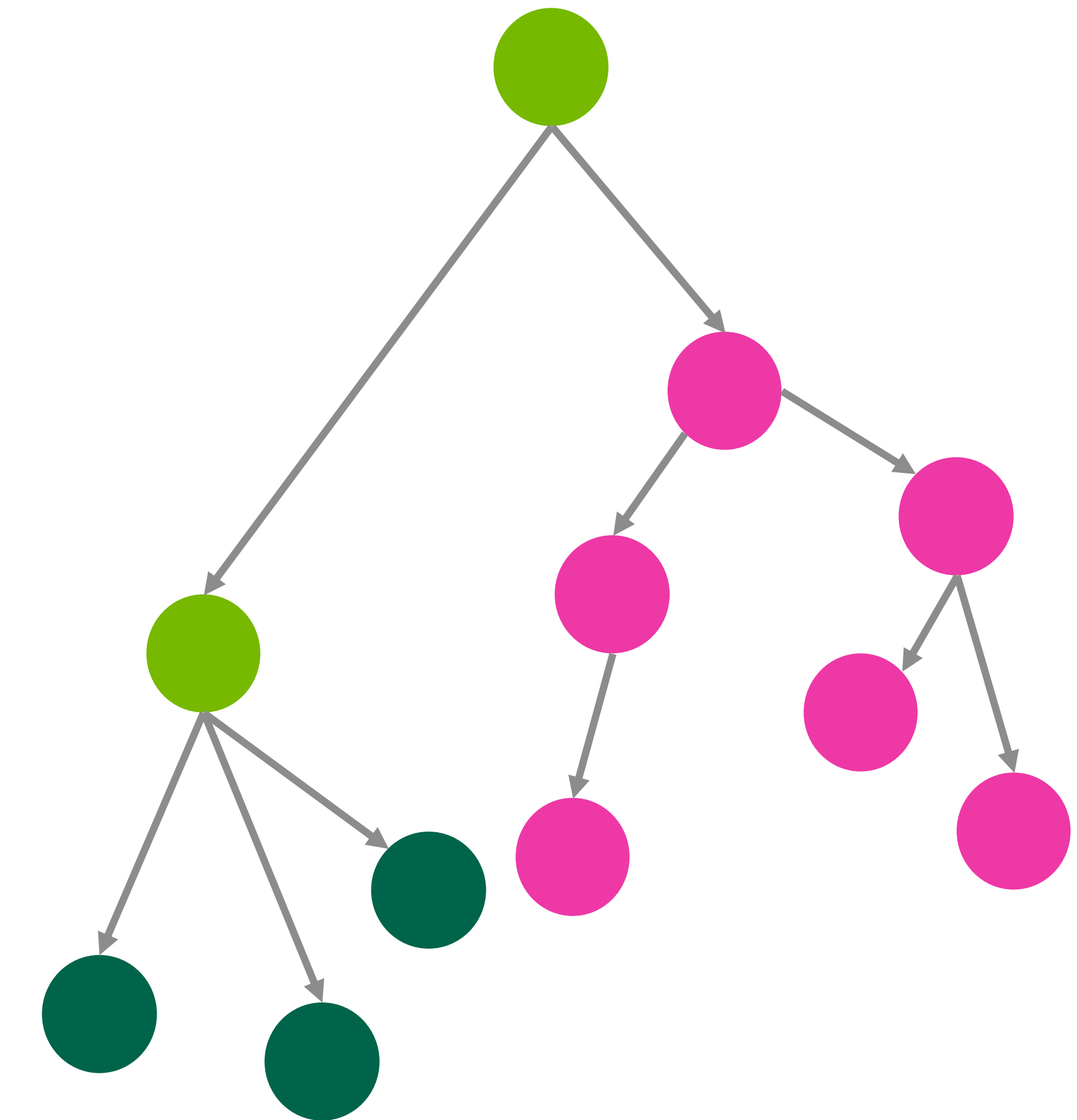
Triplets

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gas exploration", "ACTIVITY" ],  
[ "Royal Dutch Shell", "COMP", "Cut", "Spending on oil  
and gas exploration", "ACTIVITY" ],  
[ "Total", "COMP", "Cut", "Spending on oil and gas  
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[ "World's top oil companies", "ORG", "Hesitate",  
"Accelerate the search for new resources", "ACTIVITY" ],  
[ "Consultancy Wood Mackenzie", "ORG", "Estimate",  
"Global investment in exploration", "ECON_INDICATOR"  
],  
[ "Global investment in exploration",  
"ECON_INDICATOR", "Reach", "$37 billion", "VALUE" ],  
[ "Global investment in exploration",  
"ECON_INDICATOR", "Decrease", "7 percent",  
PERCENTAGE"]
```


Advanced Knowledge Graph Query Strategies

Knowledge Graphs can Increase RAG Accuracy

- Search beyond A single hop/node, by doing multi-hop expansion through breadth-first search and retrieve all relevant/connected nodes.
- Multi-hop provides far greater context about connectivity of various entities/relations compared to a single hop or semantic search.
- BFS/DFS are greatly accelerated on GPUs through implementations in cuGraph .
- How does project x relate to project y
- Multi-step LLM calls/retrieval, and final answer composition, otherwise considered an agentic workflow.
- **Key insight: knowledge graphs can accelerate multi-step and multi-hop reasoning by enabling a single retrieval from the graph to provide the necessary context for the generator LLM.**

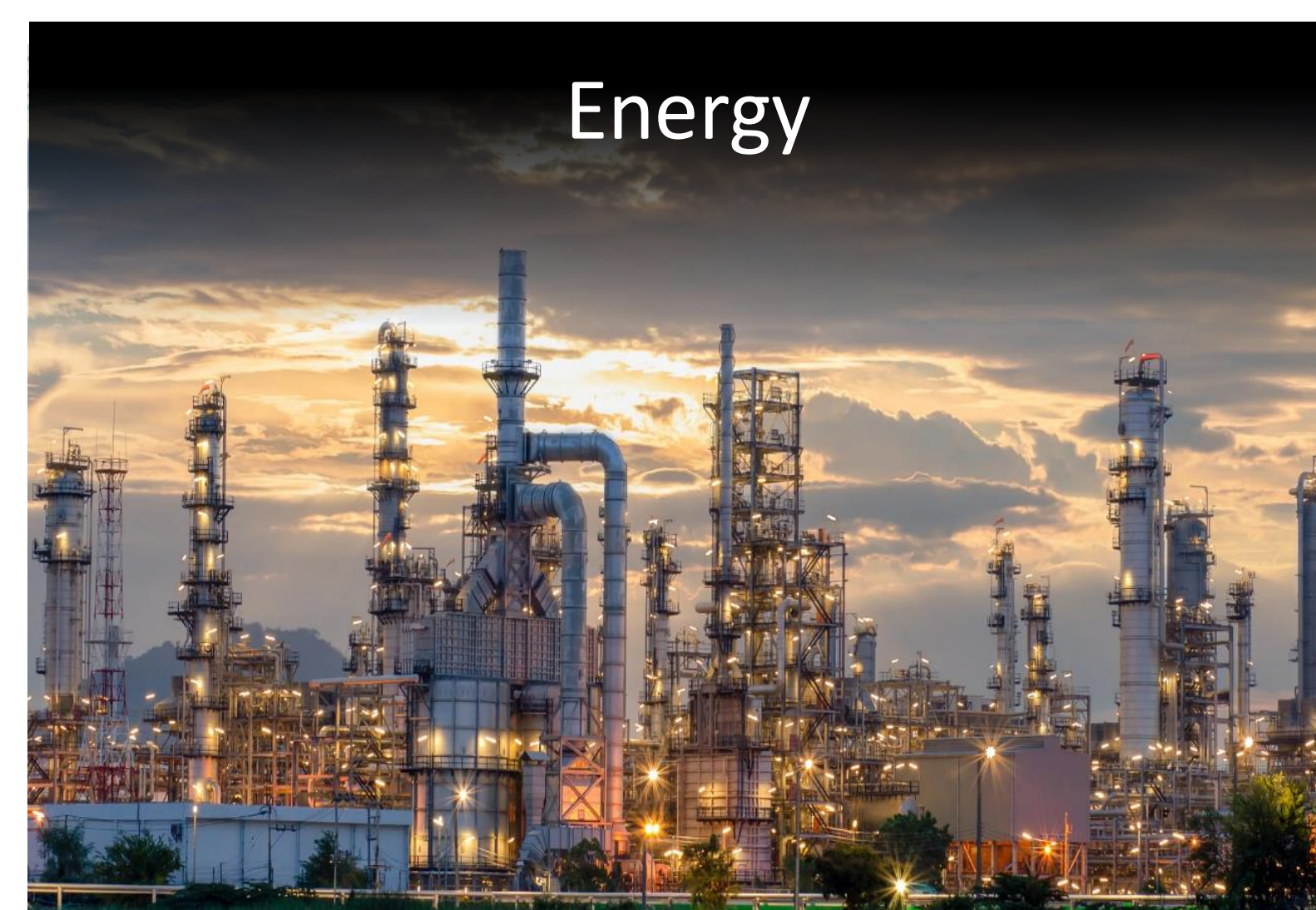




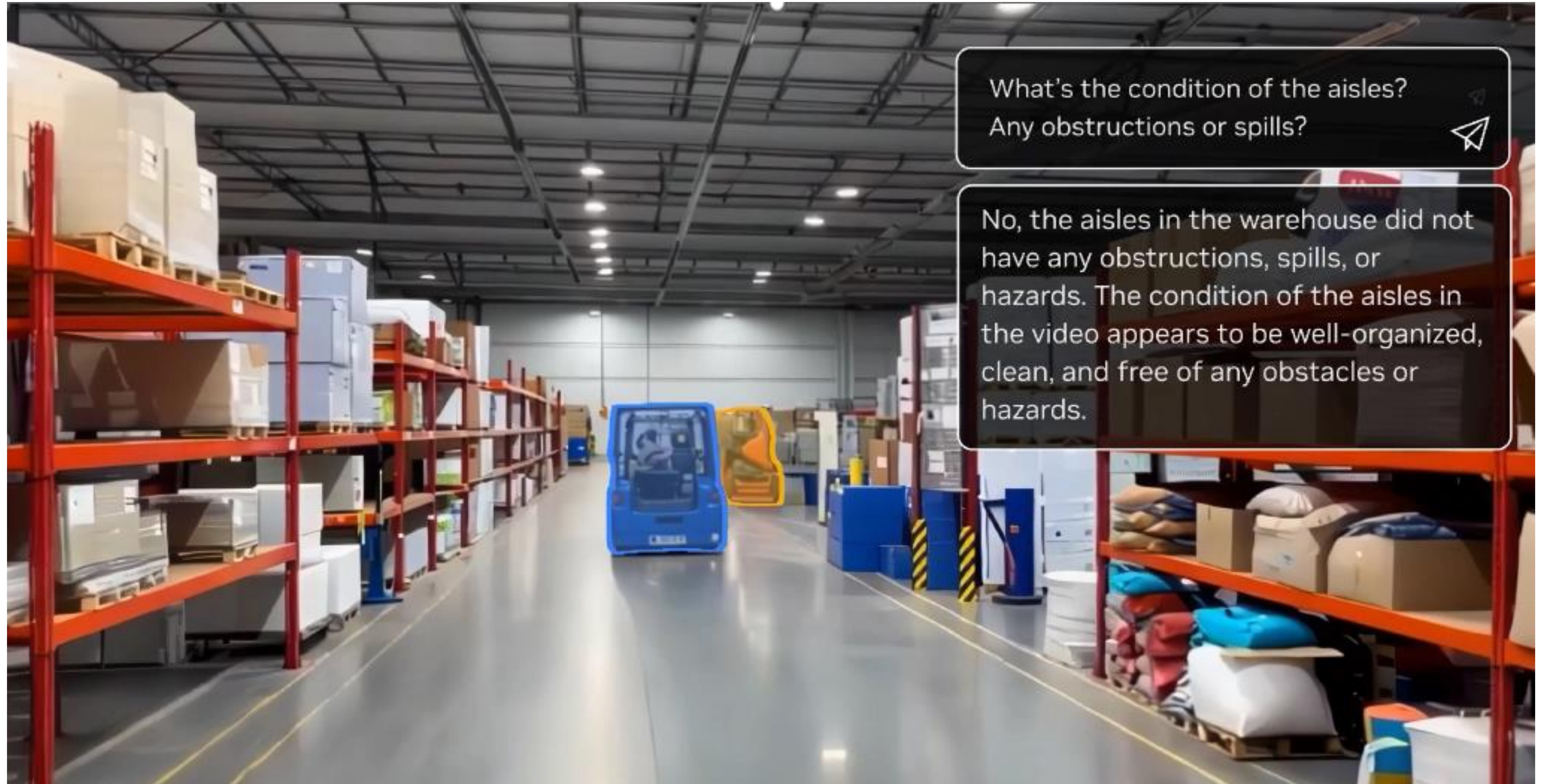
VSS Overview

Automating Important Physical Processes & Spaces

Get critical insights from massive amounts of video data



Automating Important Physical Processes & Spaces

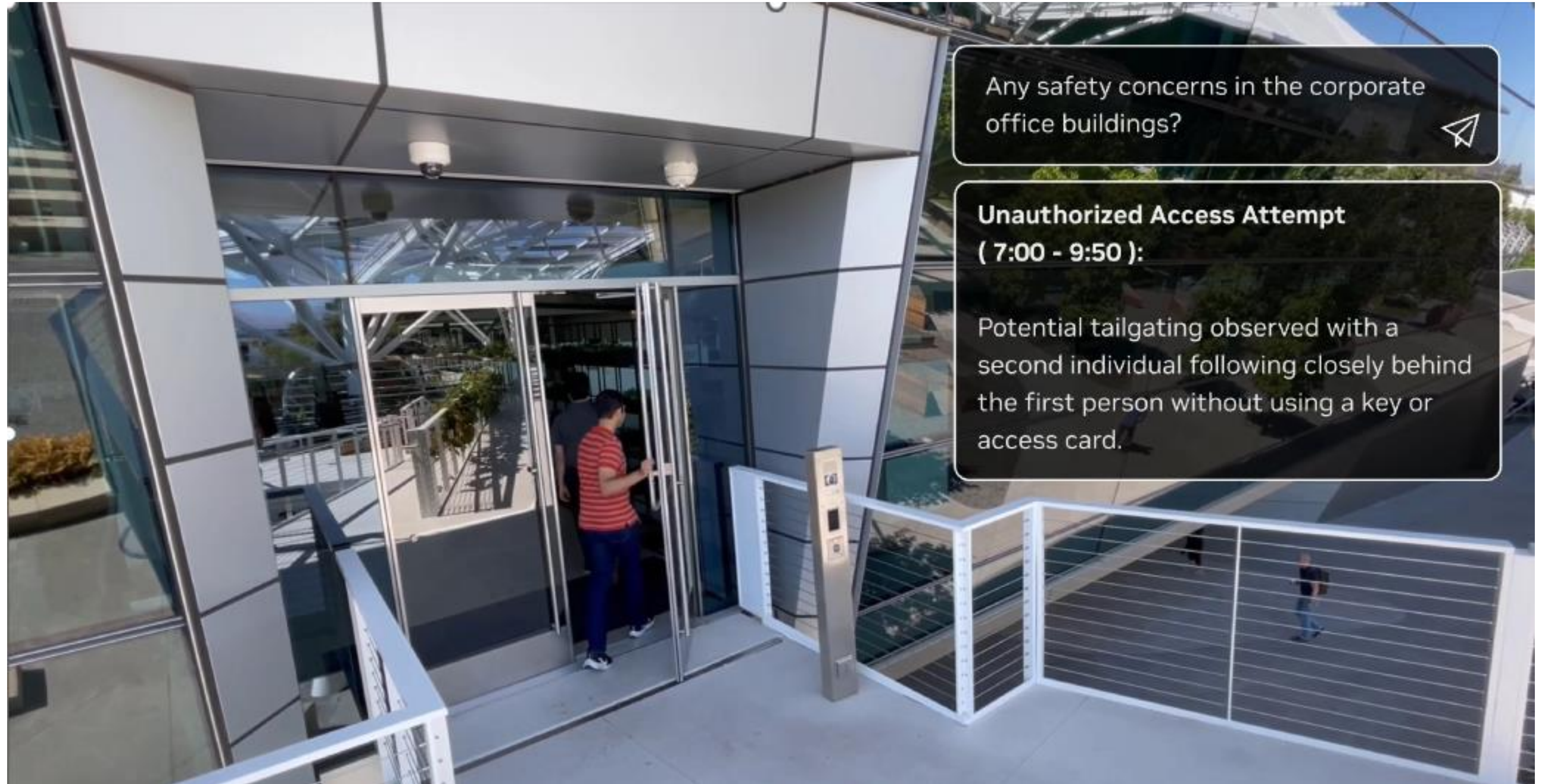


What's the condition of the aisles?
Any obstructions or spills?

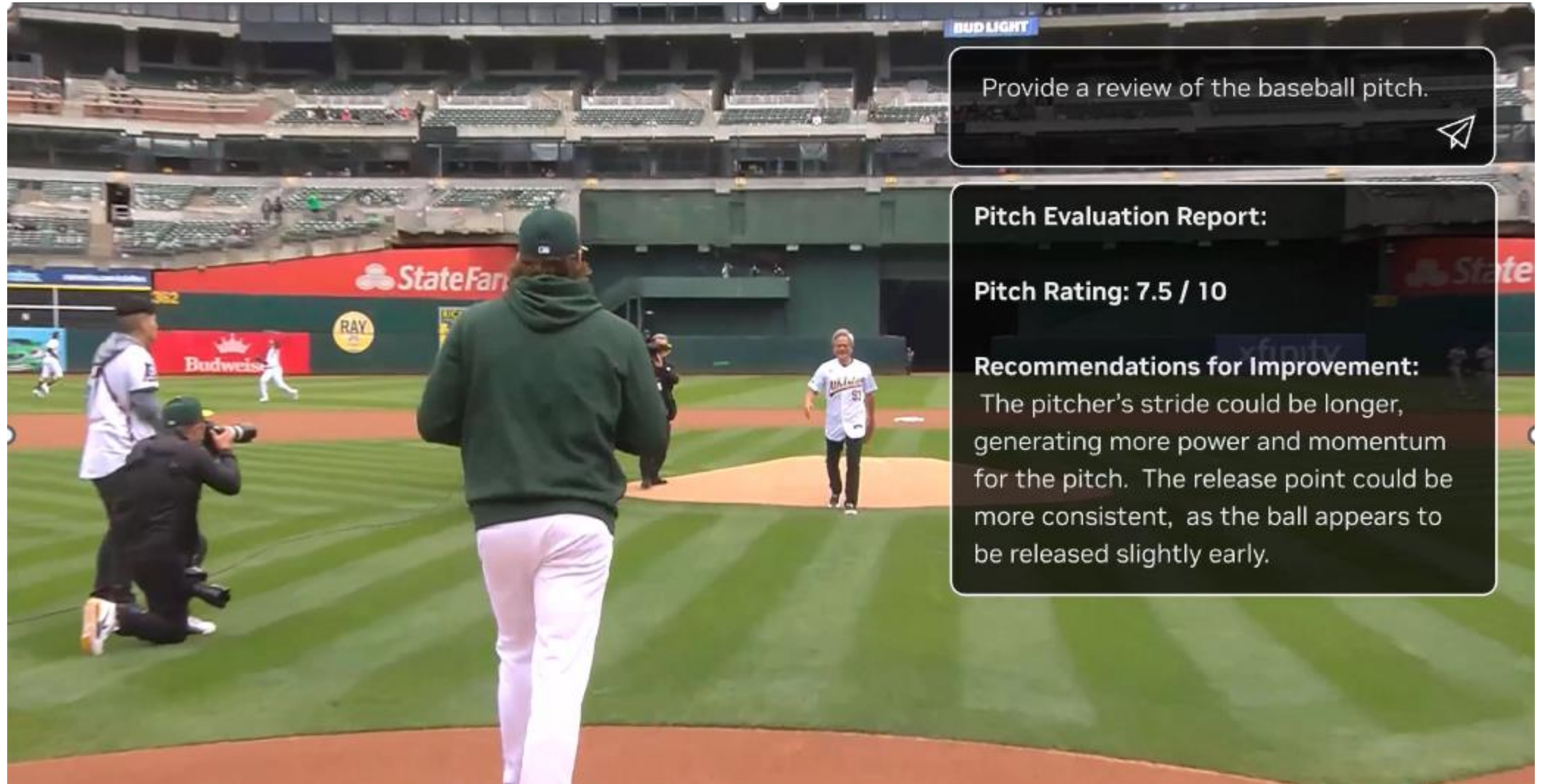


No, the aisles in the warehouse did not have any obstructions, spills, or hazards. The condition of the aisles in the video appears to be well-organized, clean, and free of any obstacles or hazards.

Automating Important Physical Processes & Spaces

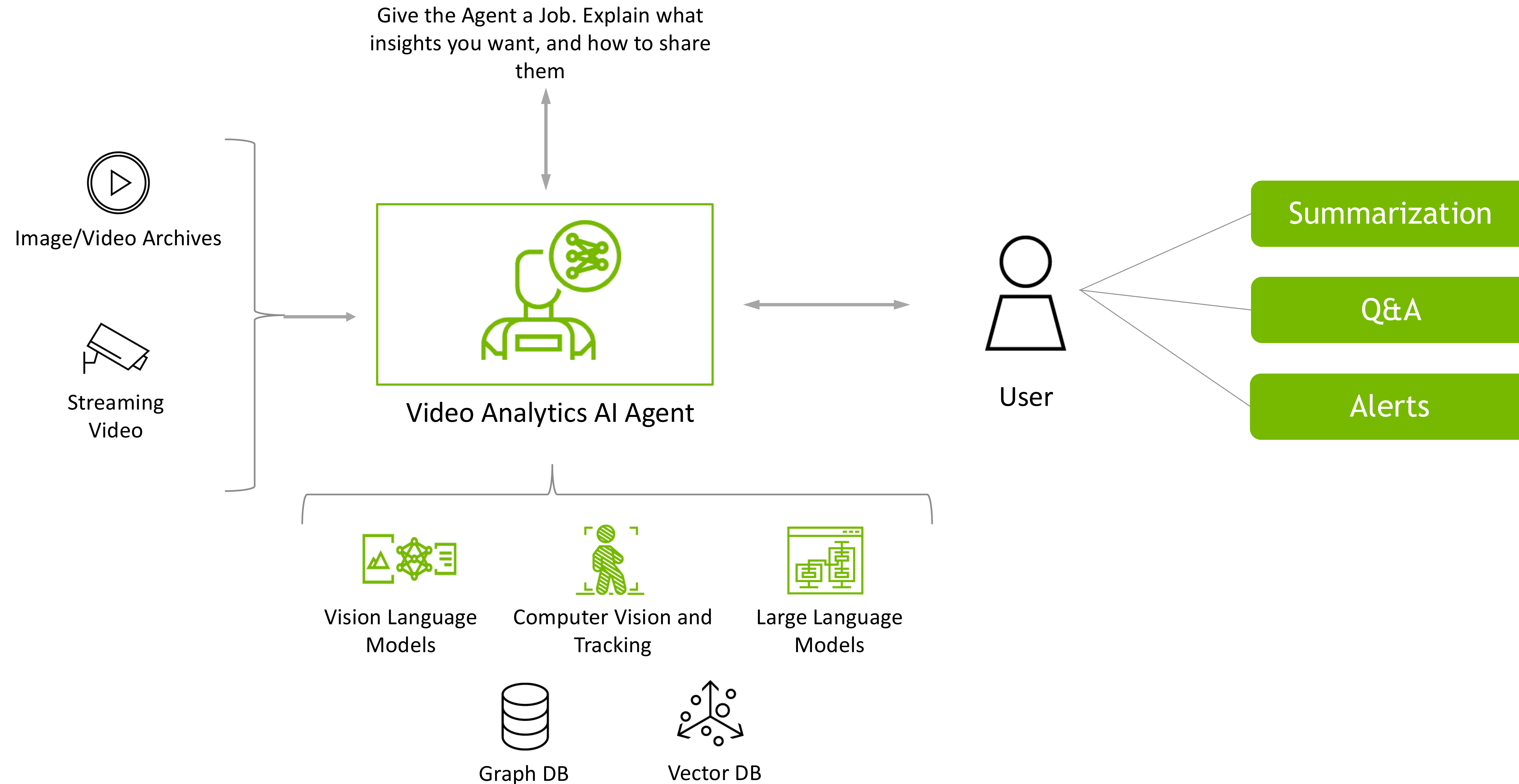


Automating Important Physical Processes & Spaces



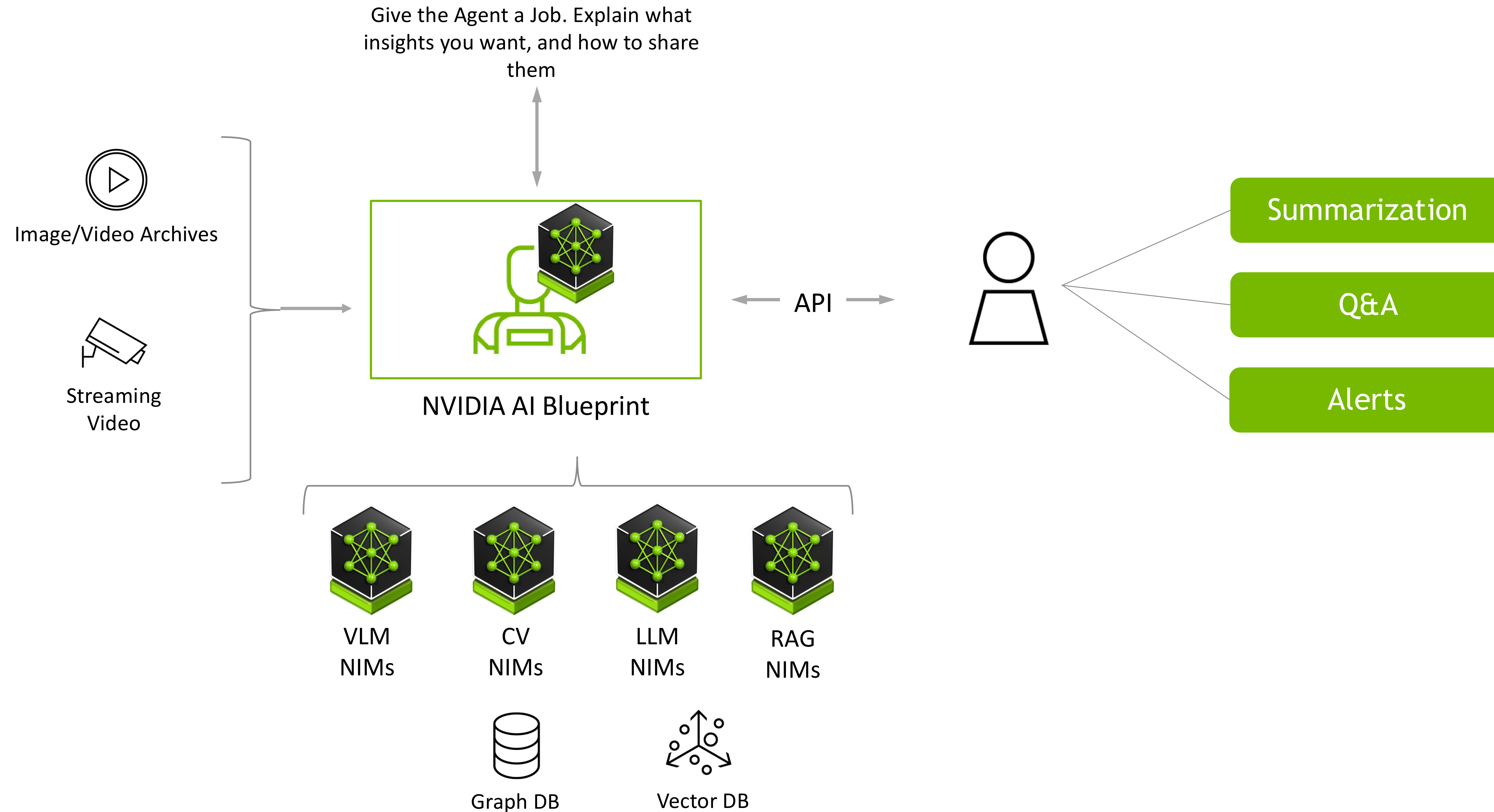
Video Analytics AI Agents – an Entirely New Class of Applications

Unlocks Knowledge & Insights from Billions of Camera Streams and Petabytes of Archived Video



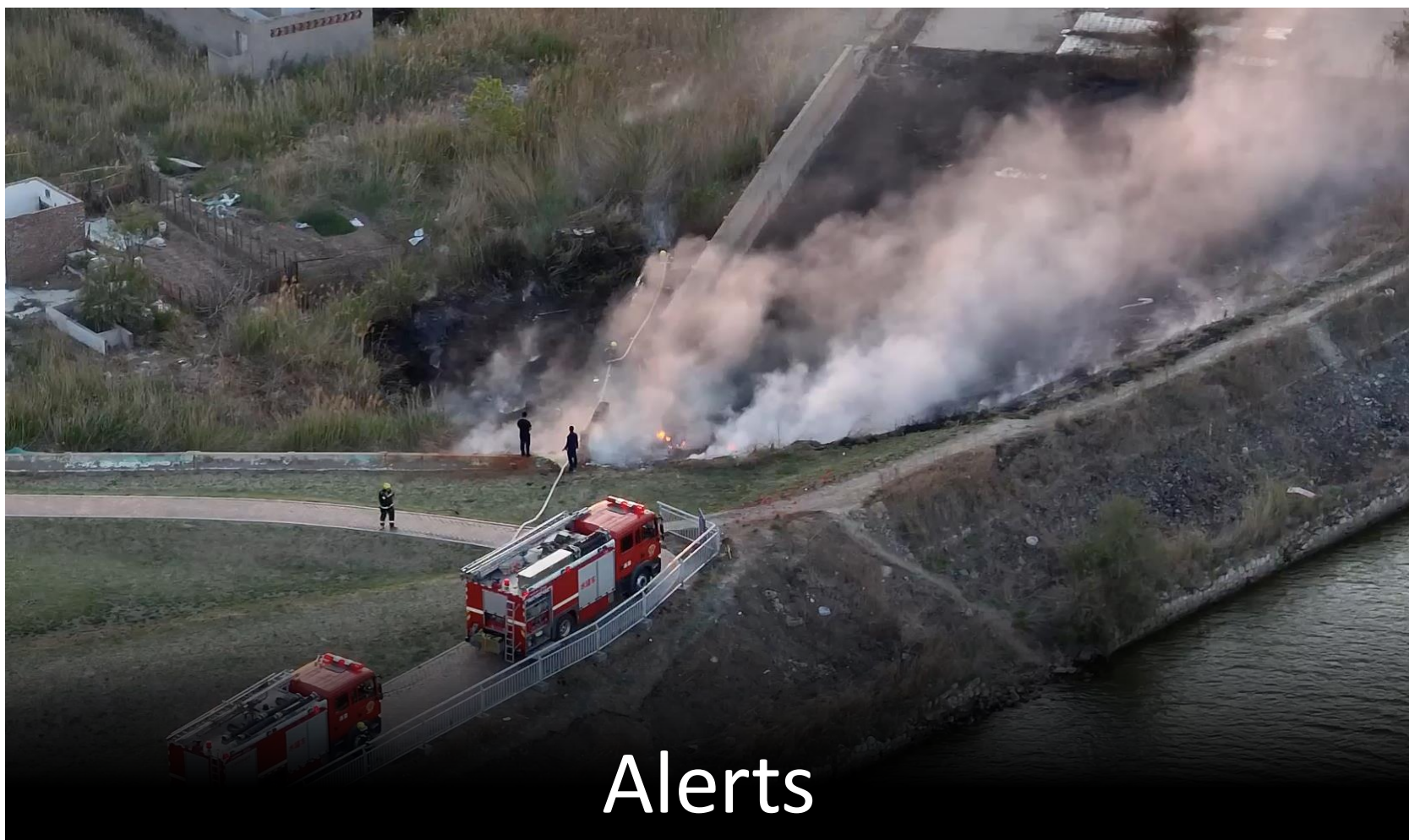
NVIDIA AI Blueprint for Video Search and Summarization

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...

NVIDIA AI Blueprint for Video Search and Summarization



NVIDIA AI Enterprise



NVIDIA NIM Models (VLM, LLM, CV, RAG)



DGX & DGX Cloud



Google Cloud



Microsoft
Azure

ORACLE

Cloud Infrastructure

DELL Technologies



Hewlett Packard
Enterprise

Lenovo



Cloud

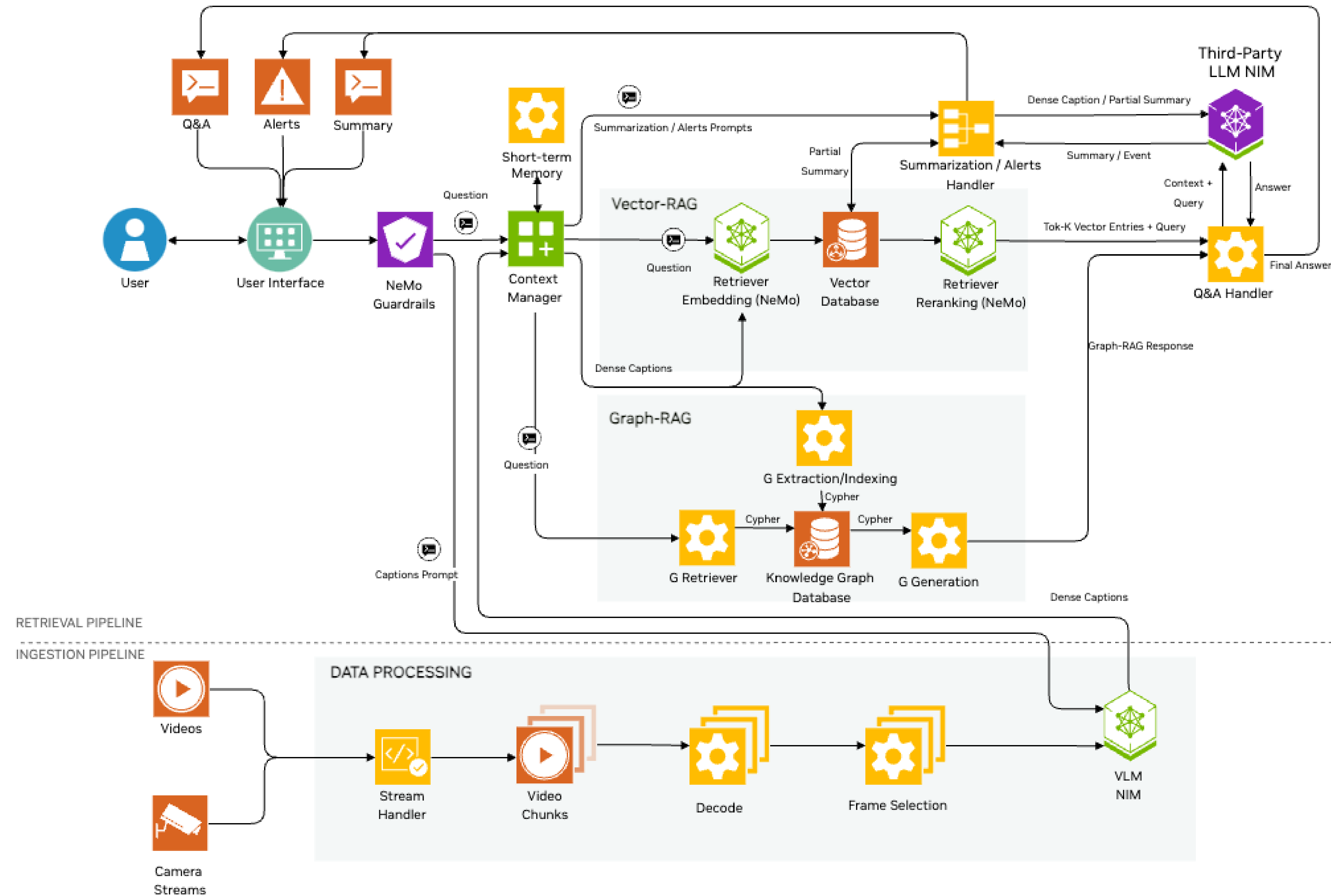


On-Prem

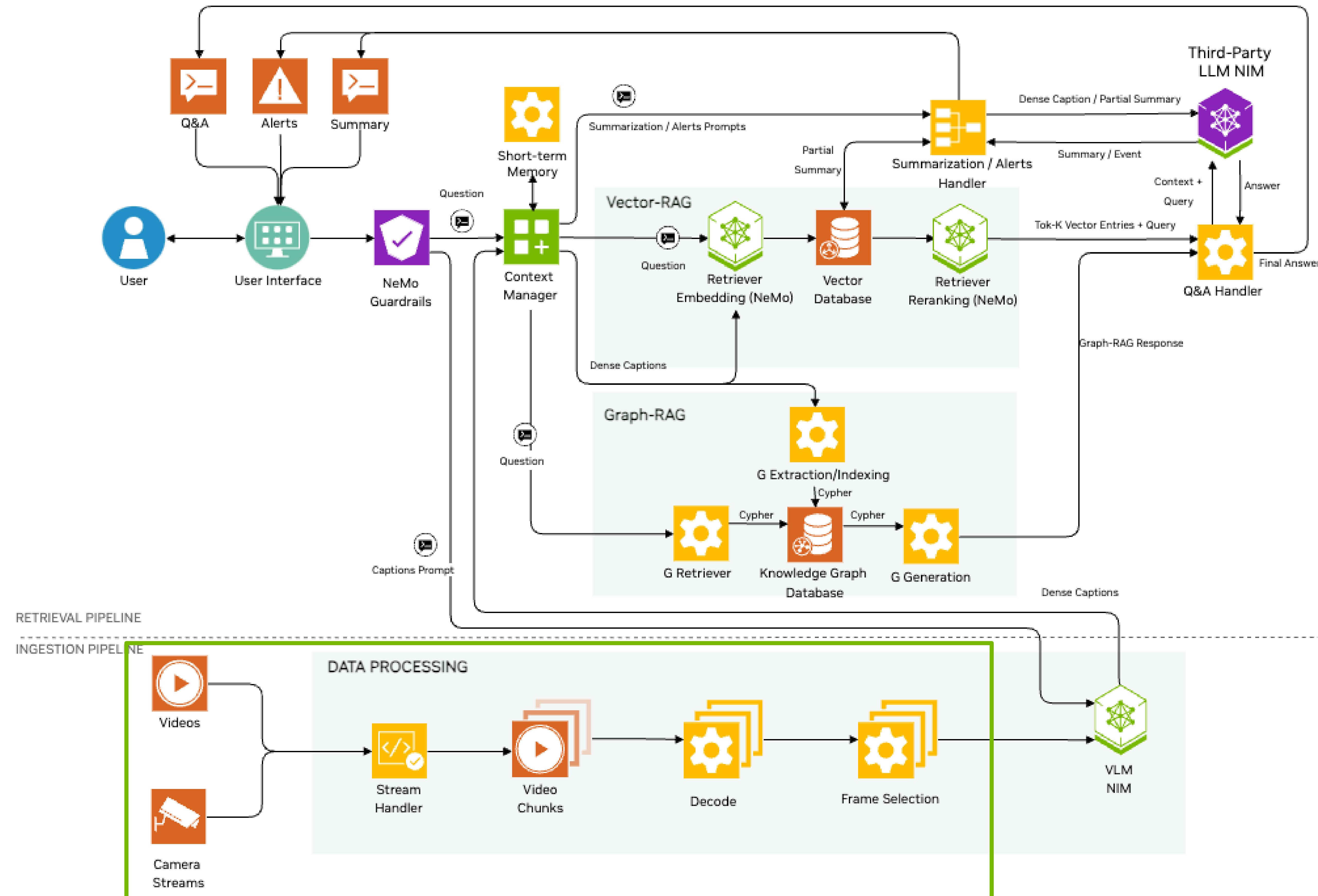


VSS Architecture

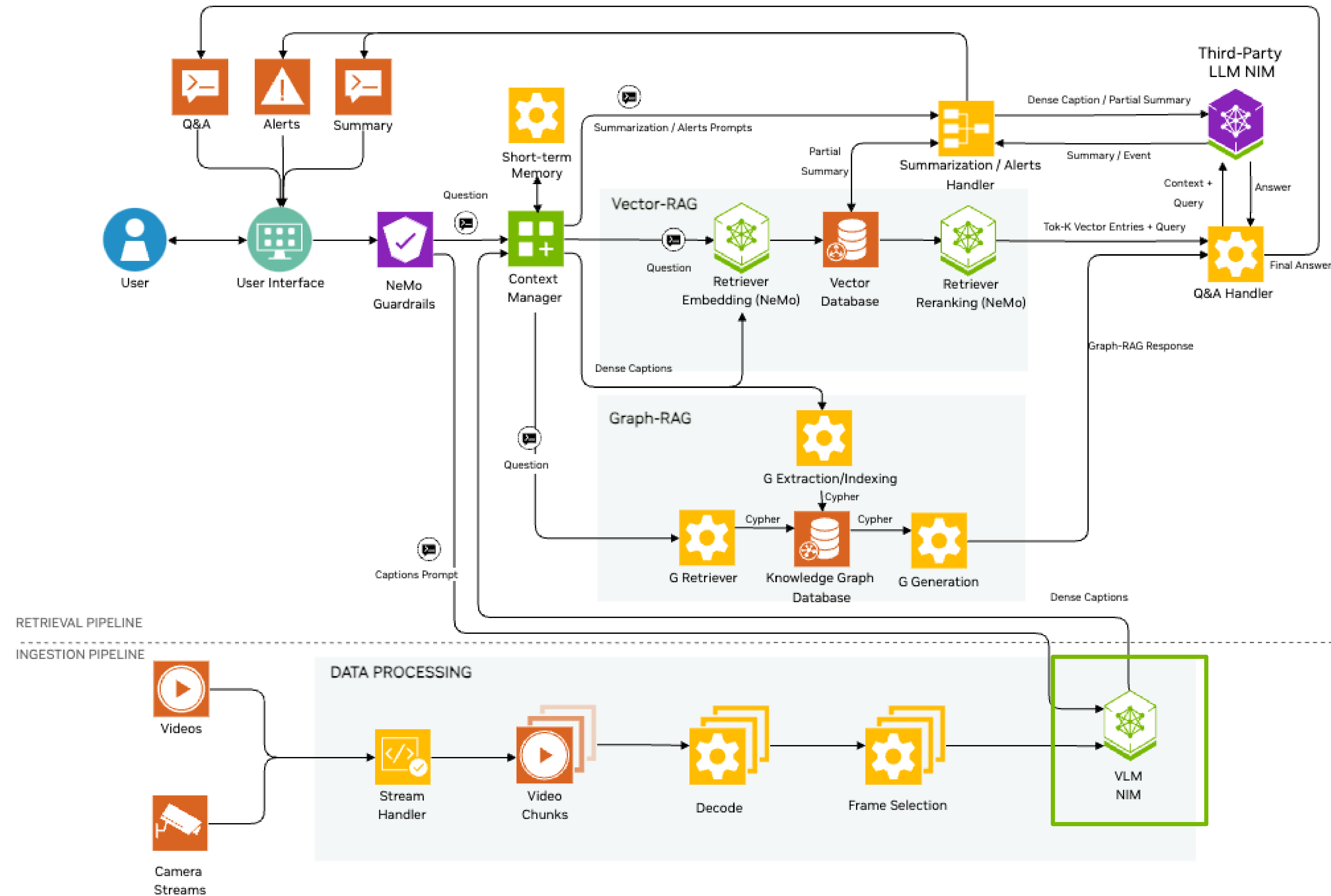
Detailed Architecture of the AI Blueprint



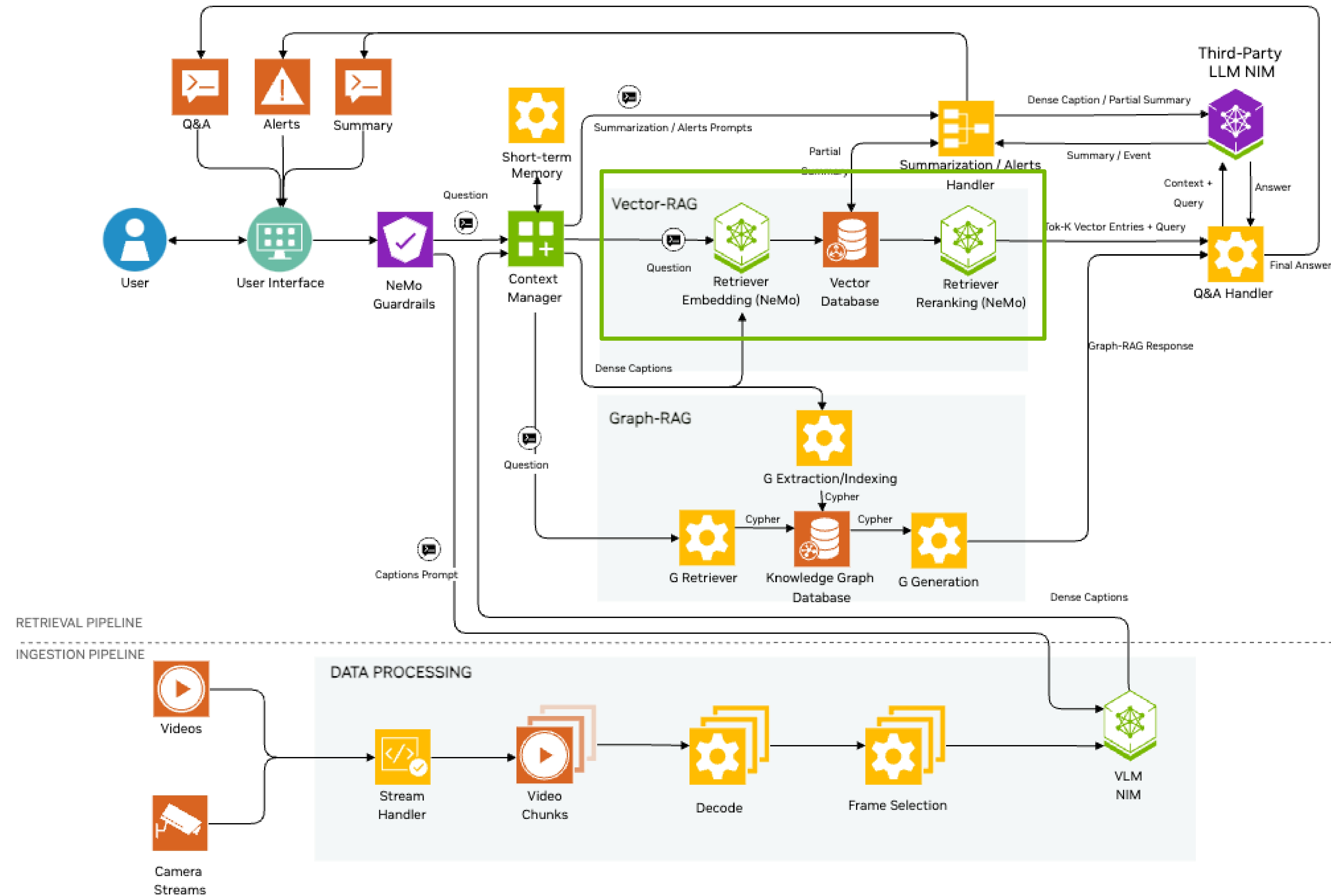
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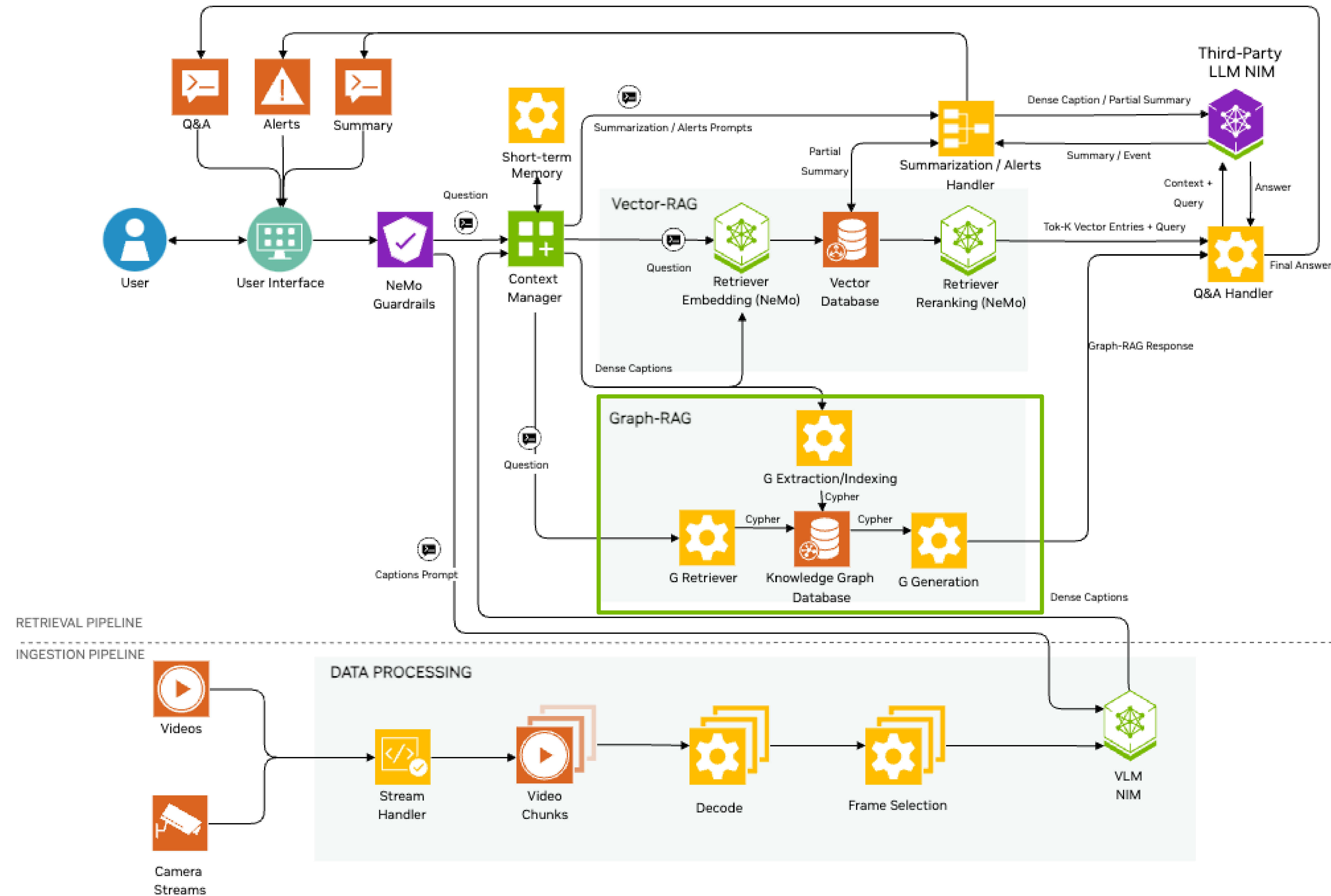
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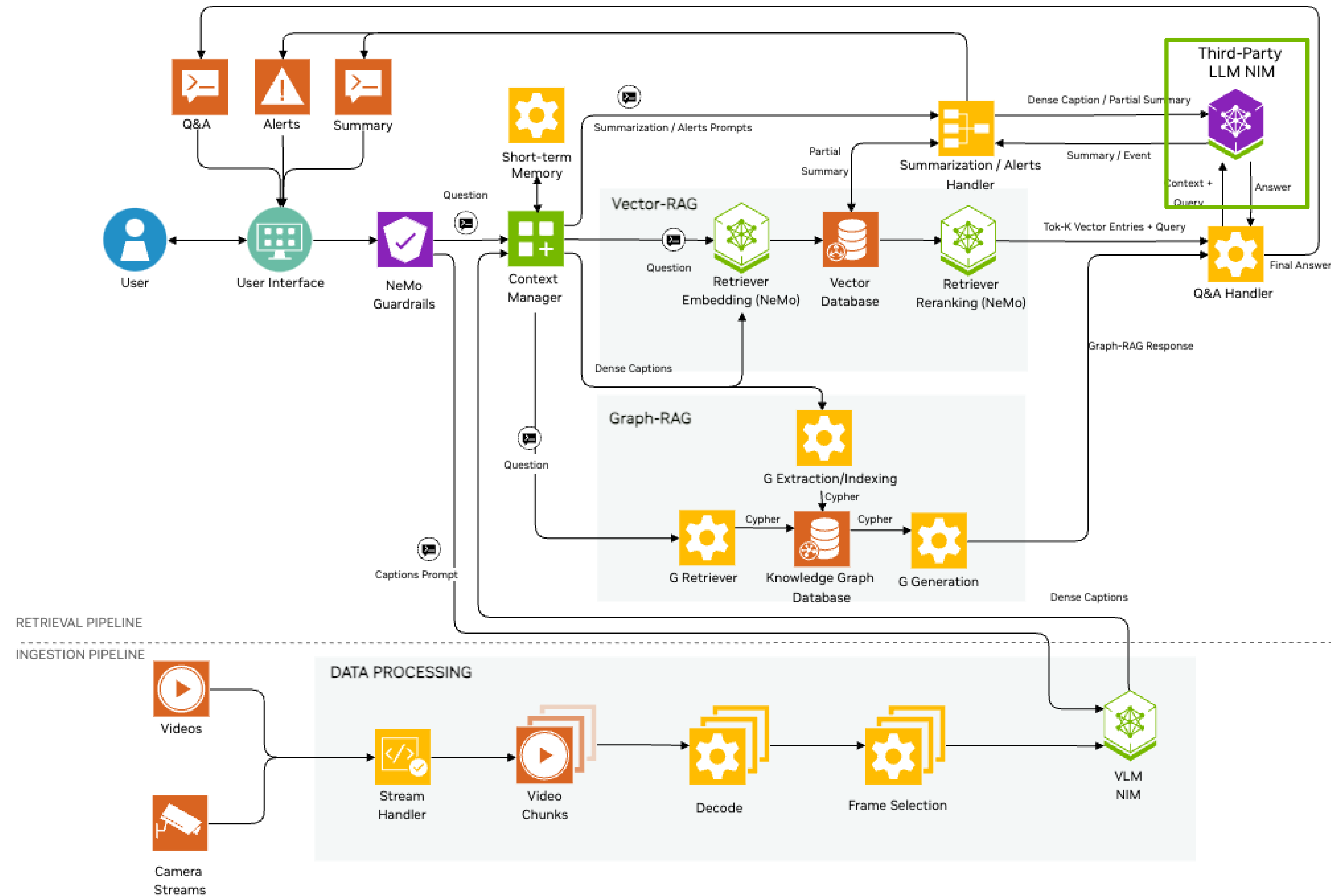
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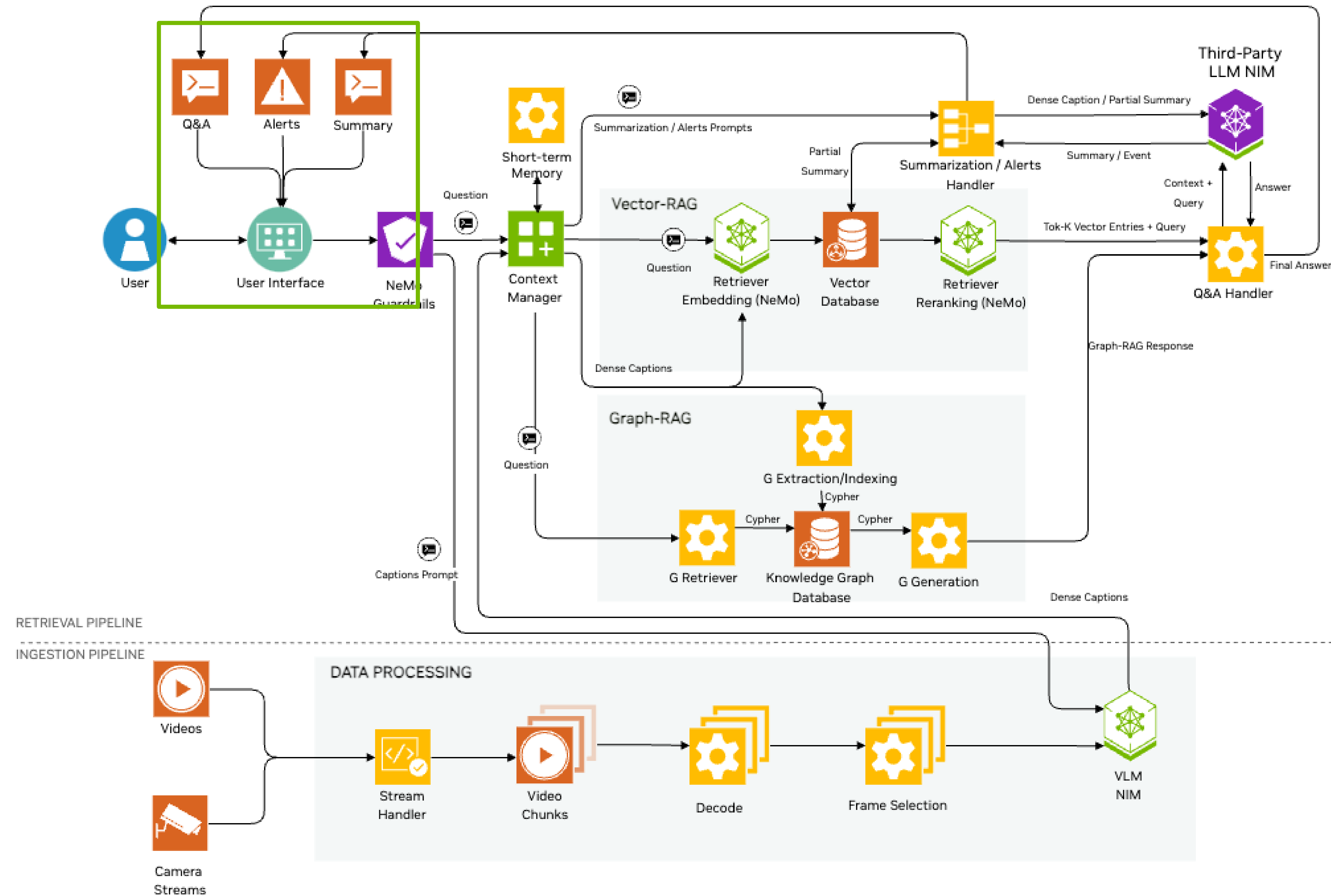
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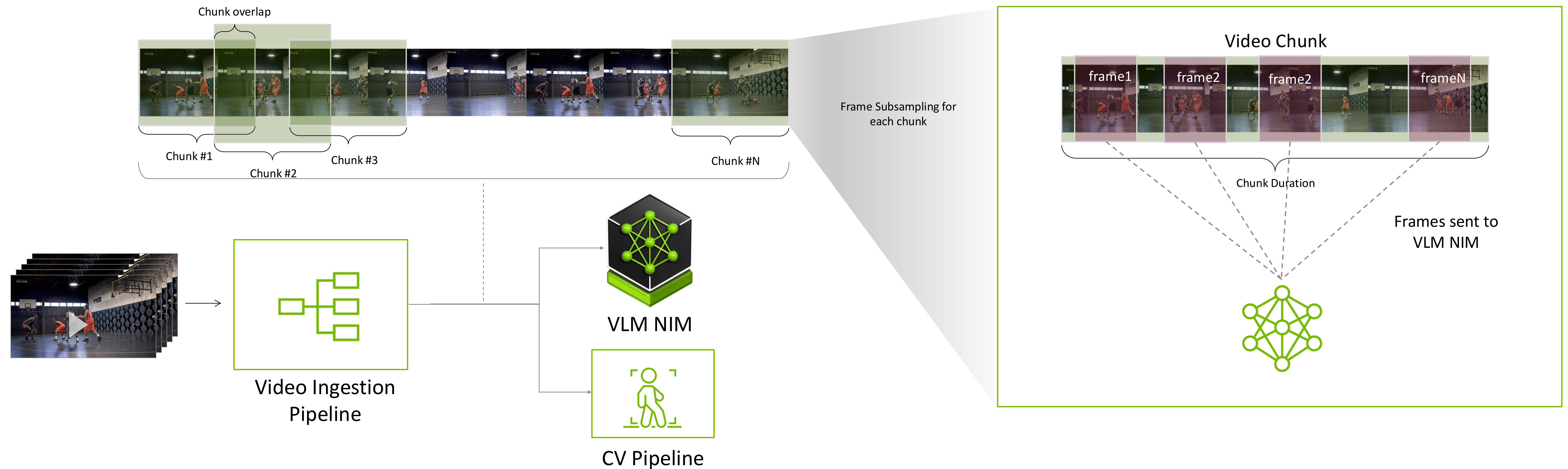
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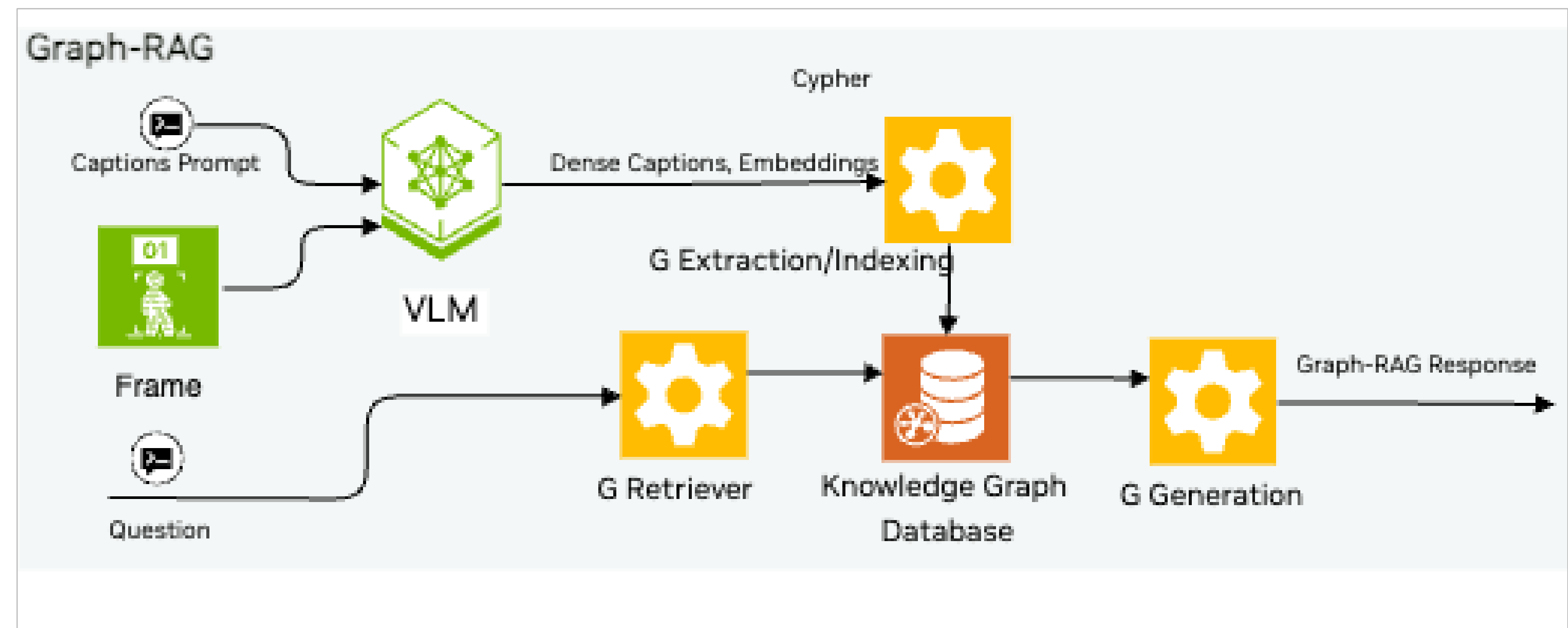
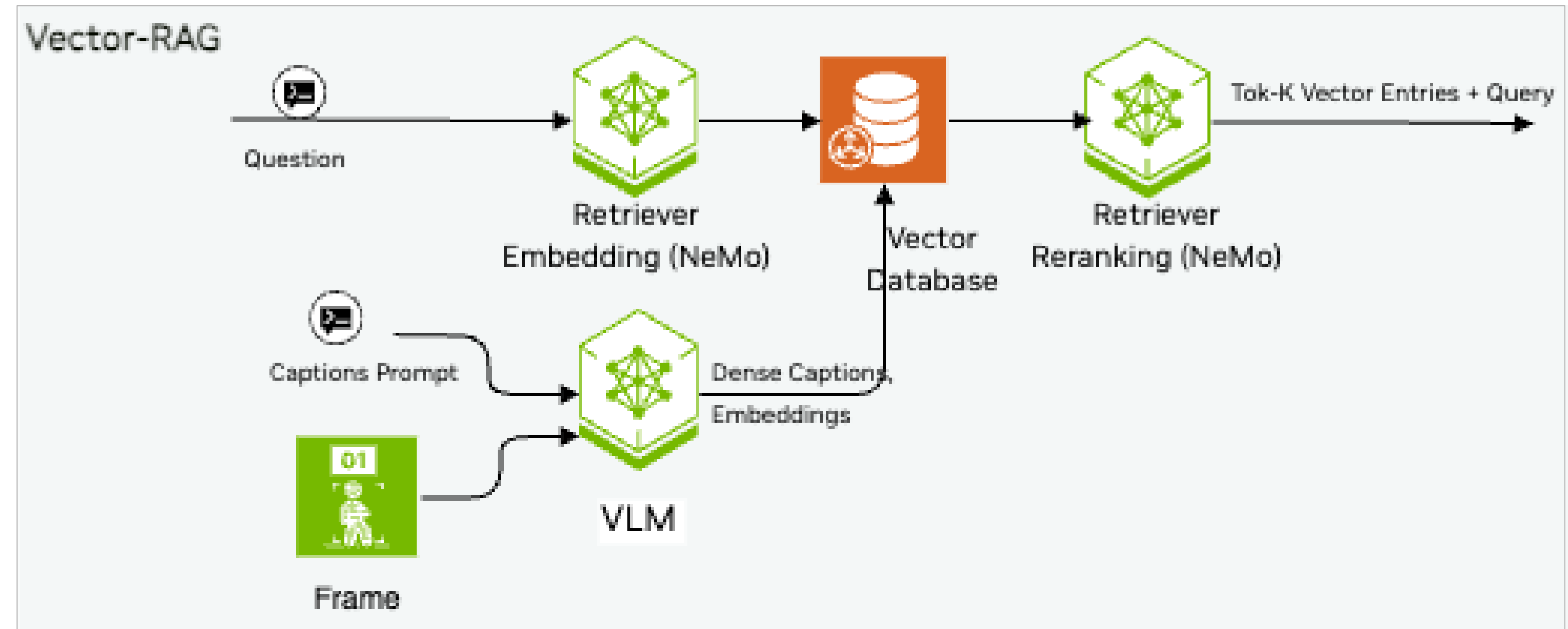


Data Ingestion & Processing



Configurable Parameter	Common Values	Impact on Accuracy	Impact of Performance
Chunk Duration	10s, 30s, 1m	Shorter chunk size works well for short actions	Higher chunk duration will yield higher performance
Chunk Overlap	10s		
Summary Duration	60s	Shorter duration can work better for more real-time update	Higher summary duration will yield higher performance
Number of frames for VLM inference	8, 10, 16, 32	More frames will yield better accuracy	More frames will reduce performance

Context Aware RAG (CA-RAG)



Context-aware multimodal processing for **video summarization and aggregation**



Coherent processing of large-scale multimodal data - **metadata, video embeddings, dense captions, timestamps**



Improved summarization quality, context retention



Aggregation algorithms, multimodal input processing and storage



API Walkthrough

Cosmos Nemotron 34B

<https://build.nvidia.com/nvidia/cosmos-nemotron-34b>

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View Examples

Video Input



Traffic Camera

Summarization Prompt

You are an intelligent traffic system. You must monitor and take note of all traffic related events. Start each event description with a start and end time stamp of the event.

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Reset

Launch Agent



Summarize the media

To begin chatting with the AI, start by clicking Launch Agent.

Cosmos Nemotron 34B

<https://build.nvidia.com/nvidia/cosmos-nemotron-34b>

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Video Input  [View Examples](#) AI Agent Output 

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Traffic Camera  mod

Summarization Prompt

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Police response and Fire Truck (100.0-130.43)
A police car with flashing lights is positioned in the middle of the intersection, blocking the red car and yellow car.
A fire truck approaches the intersection from the right and maneuvers around the stopped vehicles.

Type text here...

[Reset](#) [Launch Agent](#) [Send](#)

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View Examples

Video Input ⓘ



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Reset

Launch Agent

AI Agent Output ⓘ

waits at the top right before entering the intersection.

Turning Vehicles (50.0-69.0)
A yellow bus makes a right turn at the intersection, followed by a blue car making a similar turn.
A yellow car waits at the intersection, positioned to go straight or turn left.
A yellow car approaches from the left, stops at the stop line, and proceeds through the intersection, moving straight across and exiting the frame on the right side.

Incident and Collision (71.0-99.0)
A red vehicle approaches from the right, stops at the stop line, and remains stopped while a yellow vehicle passes it and continues through the intersection.
A red car and a yellow car collide at the intersection, and both vehicles remain stationary in the middle of the intersection, obstructing traffic.
A police car arrives on the scene with its lights flashing, stopping near the collision to respond to the incident.

Police Response and Fire Truck (100.0-130.43)
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Type text here...

Send

Cosmos Nemotron 34B

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View Examples

Video Input ⓘ



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Reset Launch Agent Send

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Video Input ⓘ



Traffic Camera

View Examples

Here are some suggestions for follow up questions

Were there any pedestrians near the car crash?

Were there any car crashes or vehicle collisions?

Summarization Prompt

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ResetLaunch Agent

AI Agent Output ⓘ

A fire truck approaches the intersection from the right and maneuvers around the stopped vehicles. The fire truck successfully navigates past the police car and continues through the intersection. The scene remains unchanged, with the vehicles stationary, possibly due to the incident.

Here are some suggestions for follow up questions.

Were there any pedestrians near the car crash?

Were there any car crashes or vehicle collisions?

Were there any pedestrians near the car crash?

No

Type text here...

Send

Build Industrial-Strength Visual AI Agents

NVIDIA AI Blueprint for video search and summarization



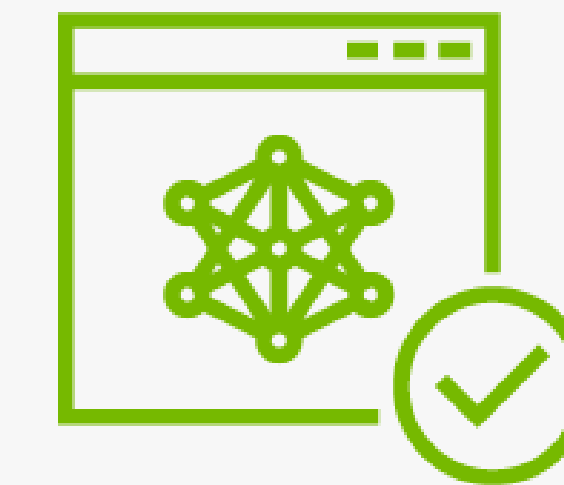
ACCELERATED TIME TO MARKET

- Pre-built workflows
- Production-ready models
- Easy to use APIs



COST EFFECTIVE AT SCALE

- Efficient data ingestion and processing
- High throughput with VLM and LLM NIMs
- Optimized architecture



HIGH ACCURACY & CUSTOMIZABILITY

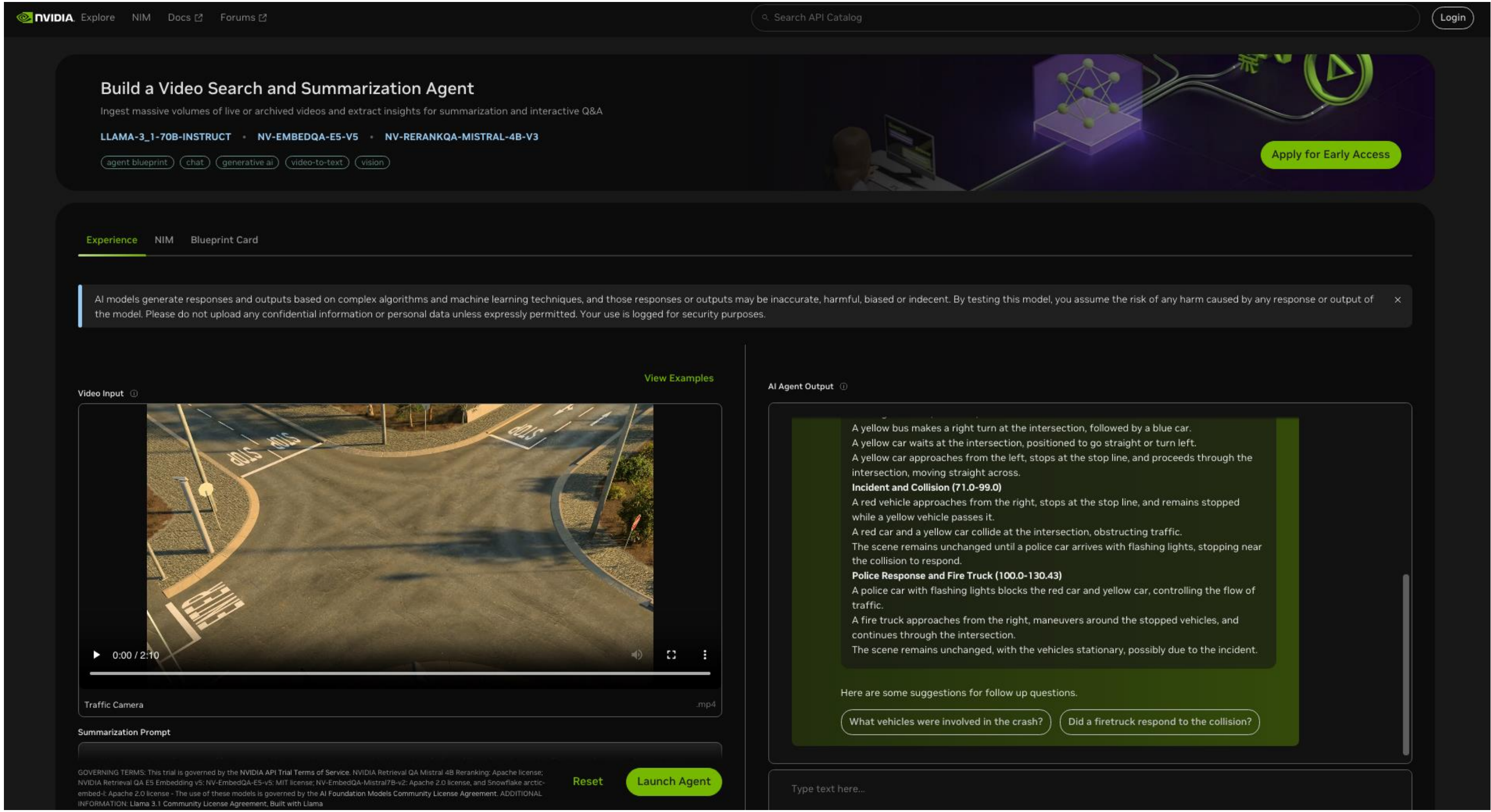
- Long context understanding with videos
- Flexible deployment from edge to any cloud
- Choice of VLM/LLM NIMs and databases
- Finetune models and prompts



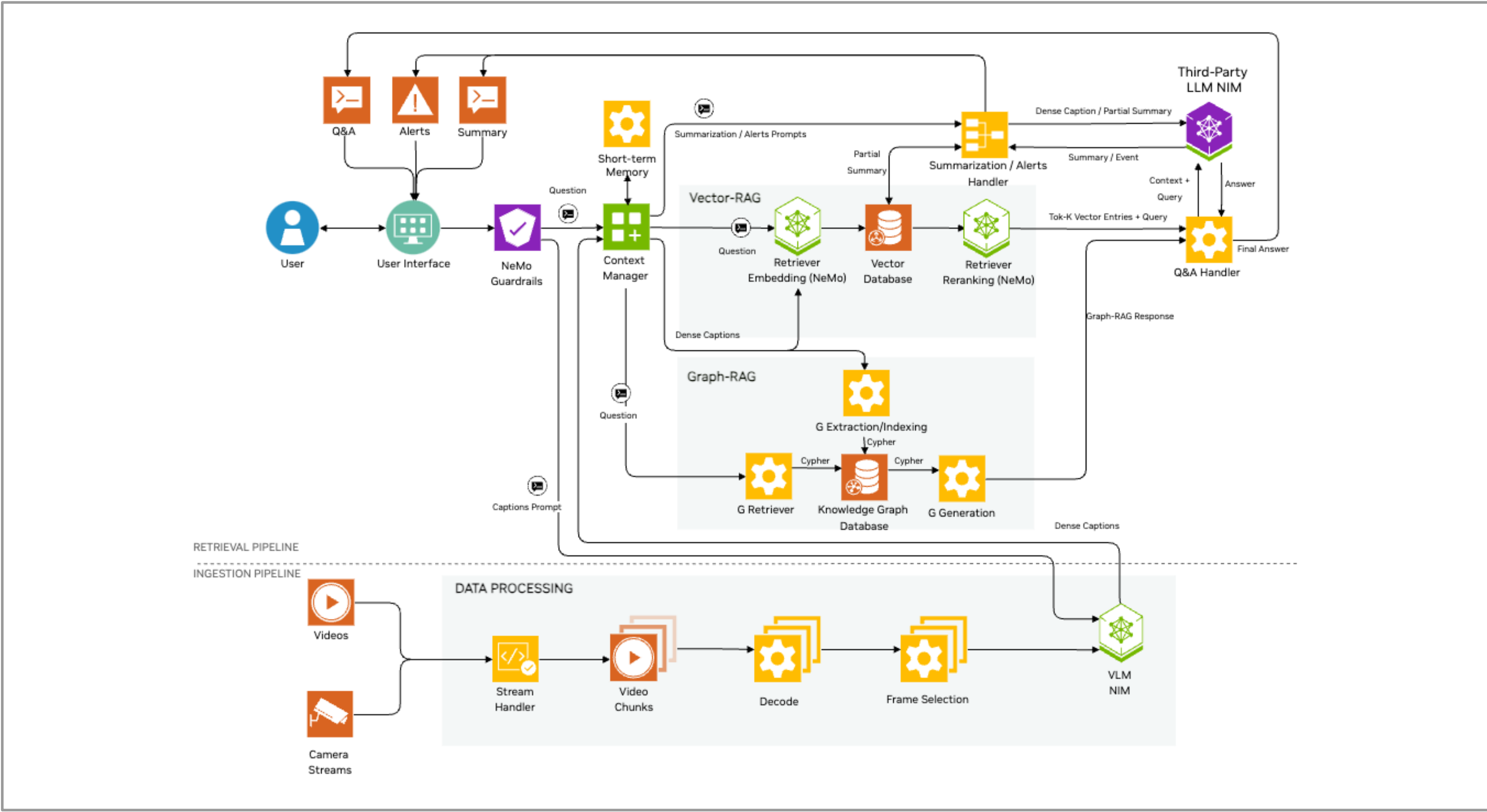
DATA PRIVACY

- Secure on-prem solution
- No need to access 3rd party services

Get Started

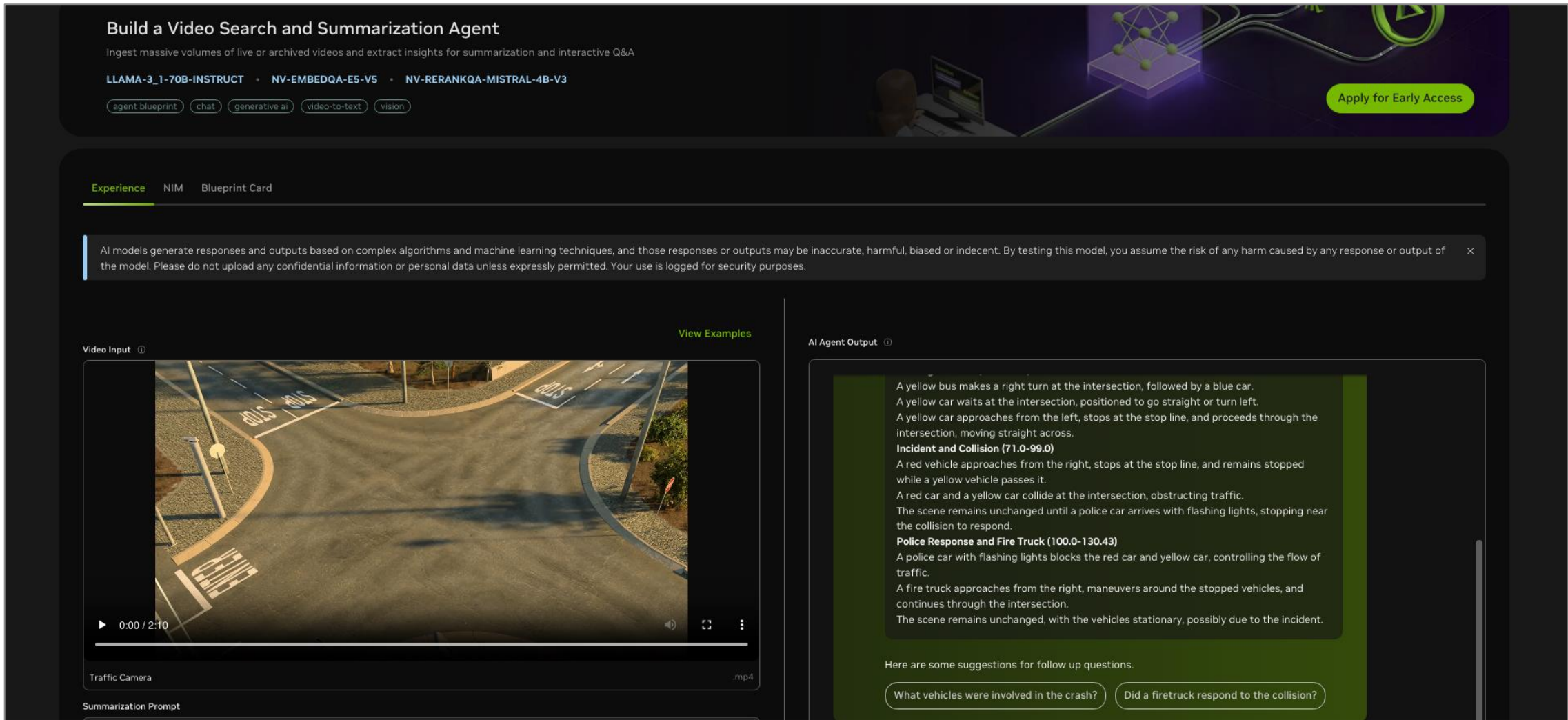


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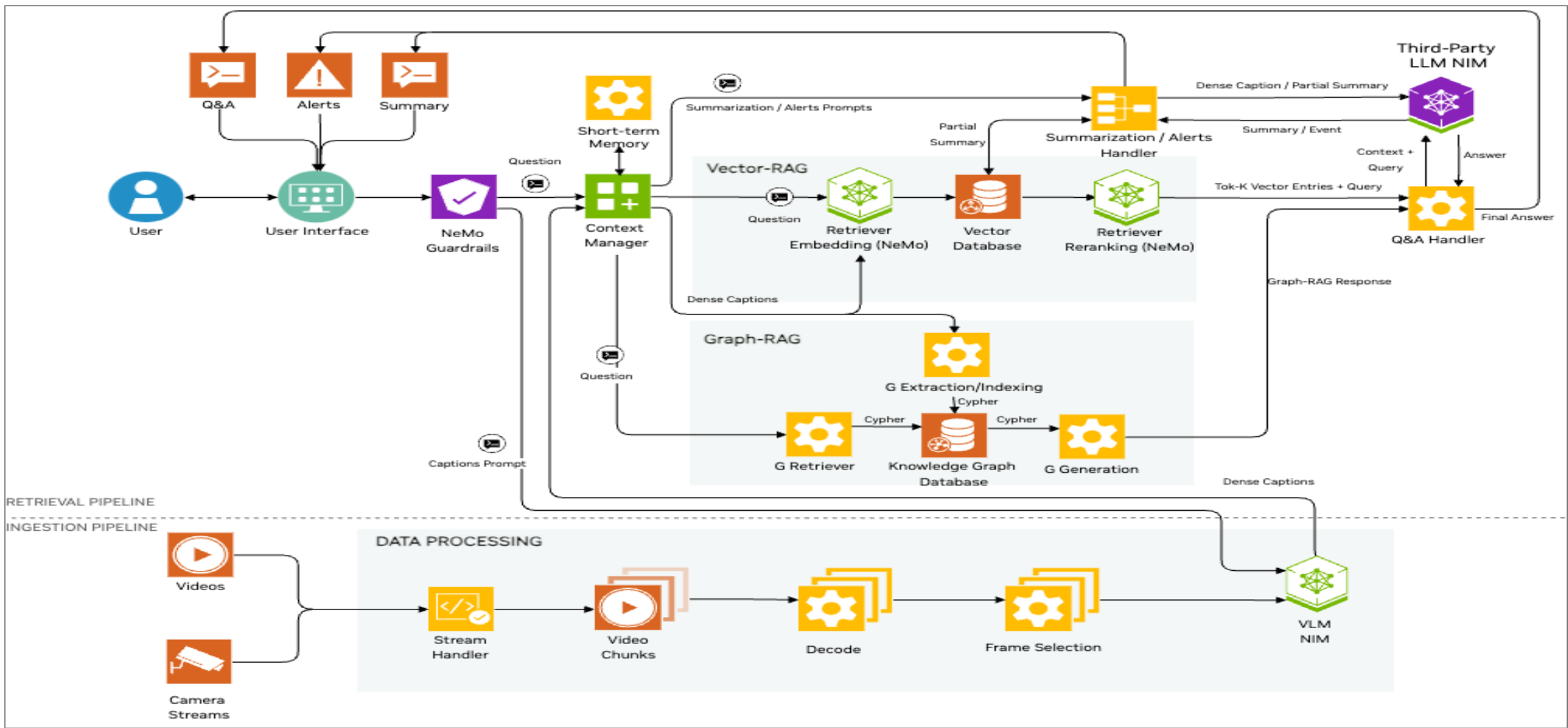


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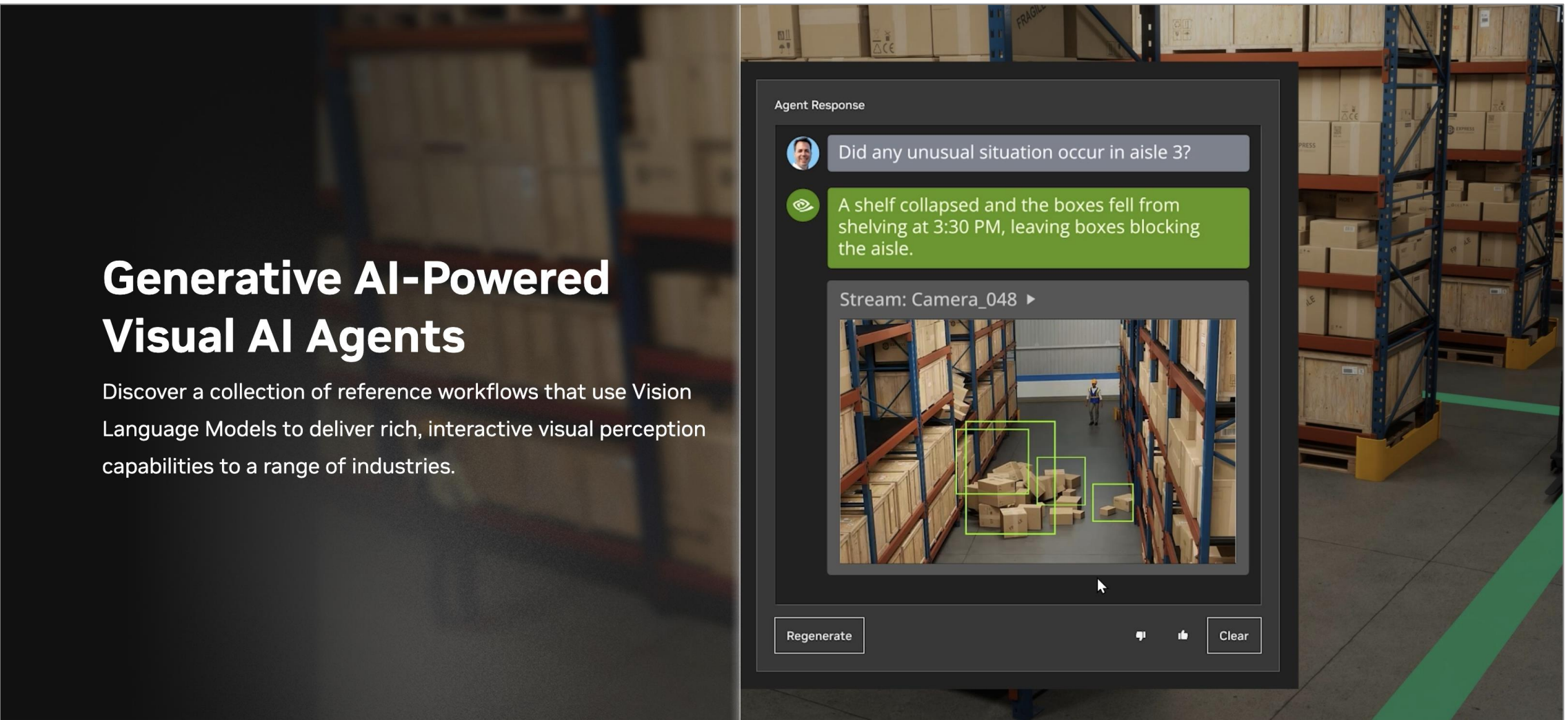
Resources



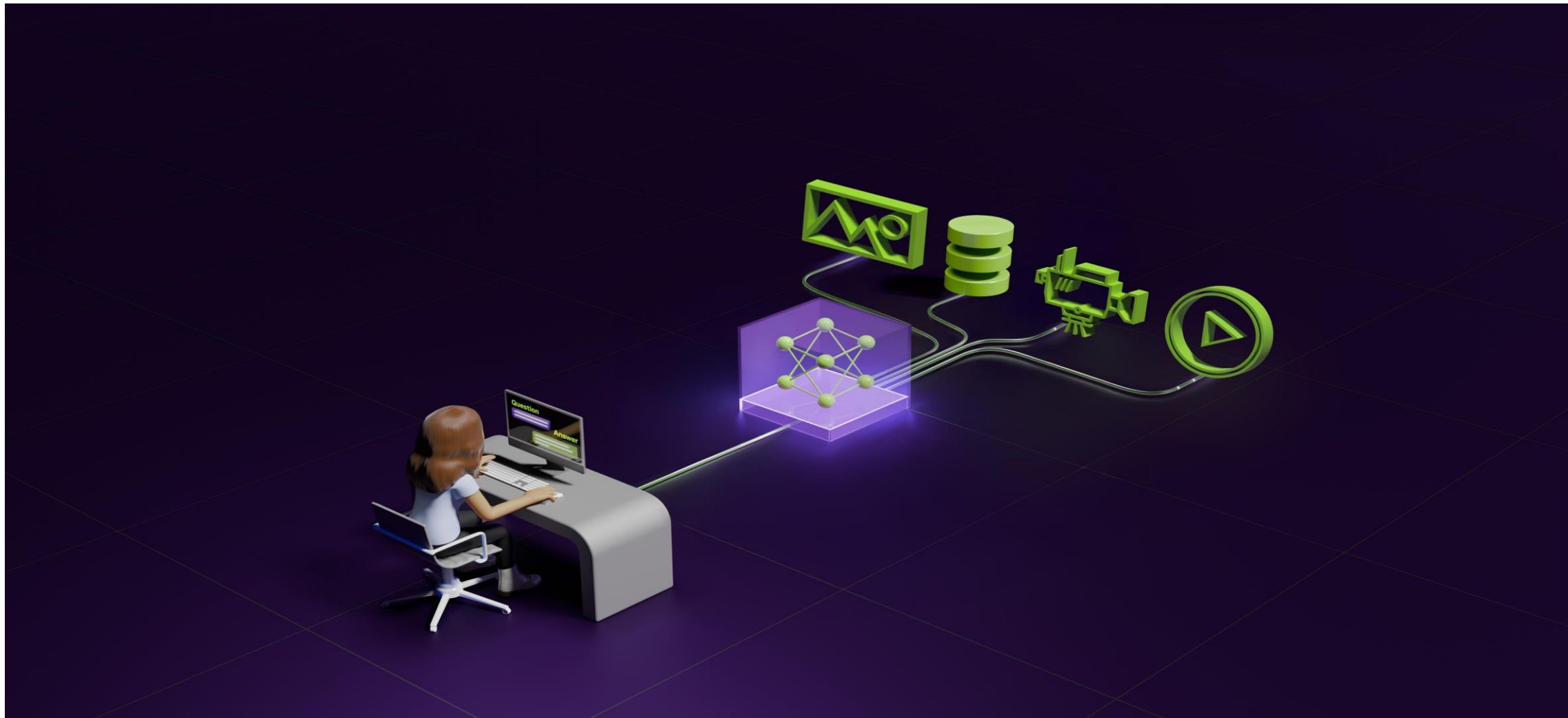
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